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Aditya University

SCHOOL OF ENGINEERING

Basic Electrical and Electronics Engineering

Unit V

BASIC ELECTRONIC CIRCUITS & DIGITAL ELECTRONICS

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Course Contents

UNIT – V

BASIC ELECTRONIC CIRCUITS

Block diagram description of a dc power supply, working of a half and full wave, bridge rectifier, filters.

DIGITAL ELECTRONICS

Overview of Number Systems, Logic gates including Universal Gates, BCD codes, Excess-3 code, Gray code, Hamming code. Truth Tables and Functionality of Logic Gates – NOT, OR, AND, NOR, NAND, XOR and XNOR. Simple combinational circuits–Half and Full Adders

INTRODUCTION

- All electronic circuits need D.C. power supply either from battery or power pack units. It may **not be economical and convenient to depend upon battery power supply**. Hence, many electronic equipment contain circuits which convert the A.C. supply voltage into D.C. voltage at the required level.

- The unit containing these circuits is called the **Linear Mode Power Supply (LPS).** In the absence of A.C. mains supply, the D.C. supply from battery can be used. Also, in certain applications, D.C. to D.C. conversion is required. Such a power supply unit that converts D.C. into A.C. or D.C. is called **Switched Mode Power Supply (SMPS).**

- Linear power supply (LPS): A.C./D.C. power supply—Converter

- Switched mode power supply (SMPS):

(i) D.C./D.C. power supply—Converter

(ii) D.C./A.C. power supply—Inverter

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INRODUCTION

- An A.C./D.C. power supply converts A.C. mains (230 V, 50 Hz) into required D.C. voltages and is found in all mains operable system.
- The D.C./D.C. power supplies or D.C./A.C. converters are used in portable systems.
- D.C./A.C. power supplies or inverters are used in portable mains operable systems and as a supplement to A.C. mains in non-portable mains operable system, where a disruption in the power supply can affect the job being done by the system.

Linear Mode Power Supply

Several electrical and electronic circuits need a DC power source to operate. But we are unable to employ common DC power sources like batteries and DC generators due to several technological and financial limitations.

In this case, we employ an electronic circuit called as a **rectifier** that can transform AC power into DC power.

A rectifier's output DC power, on the other hand, **is a fluctuating DC voltage and current that could harm an electronic circuit**. Thus, to overcome the problem of variation in voltage, some voltage regulating devices are used so that we can obtain a steady DC power supply.

An electronic circuit that produces a stable DC voltage of fixed value across the load terminals irrespective of changes in the load is known as regulated power supply (RPS). Thus, the primary function of a regulated power supply is to convert an AC power into a steady DC power. **The regulated supply is sometimes called as a linear power supply.** The block diagram of regulated power supply is shown in figure (6.18).

Linear Mode Power Supply

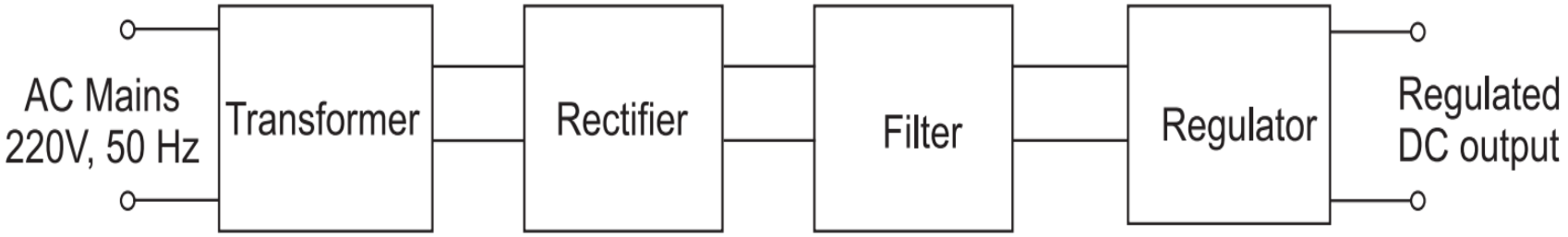


Figure (6.18): *Regulated Power Supply*

The regulated power supply consists various parts, such as

(i)transformer, (ii)rectifier, (iii) filter, and (iv) regulator

(i) Transformer: A transformer modifies the alternating current mains voltage to a required value, and its primary duty is to step up or step down the voltage. A step-down transformer, for example, is used in a transistor radio, while a step-up transformer is used in a CRT. Transformers provide separation from the power line and must be used even when voltage changes are not necessary.

Linear Mode Power Supply

(ii) Rectifier: The rectifier circuit is used to convert the input AC voltage into a DC voltage. **The input to a rectifier is AC whereas its output is pulsating DC**. Rectification process can be carried out for the entire cycle or only for half-cycle. Broadly there are two types of rectifiers such as **(i)half-wave rectifier, and (ii)fullwave rectifier**. Full-wave rectifier is used for full wave rectification and half-wave rectifier provides rectification for half-cycle.

iii) Filter: The rectifier's output is pulsating direct voltage with large ripple content and hence, the rectifier's output is undesirable. **A filter is used to provide a pure ripple-free direct voltage**. The filter circuit converts the pulsating direct voltage into a constant direct voltage with almost zero ripple content. Different types of filter circuits such as **capacitor (C) filter, choke input (L) filter, CLC filter and LC filter**.

Linear Mode Power Supply

(iv) Regulator: The output voltage may change or fluctuate due to any change in the **input ac voltage** or the **change in load** or **change in physical parameters** such as temperature of the circuit. This problem can be eliminated with the help of a regulator. A regulator maintains the output constant even when changes at the input voltage or any other changes occur. Different types of regulator circuits such as **transistor in series, Zener diode operating in Zener region and IC regulators** are available.

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Rectifiers

- Rectifier is defined as an electronic device used for converting A.C. voltage into unidirectional voltage (i.e DC Voltage).
- A rectifier utilizes unidirectional conduction device like a vacuum diode or PN Junction diode.
- Rectifiers are classified depending upon the period of conduction as
  - **Half-wave rectifier**
  - **Full-wave rectifier.**

Half Wave Rectifier

In half wave rectifier, the rectifying element i.e. diode conducts only either for positive half cycle or negative half cycle. The rectifier circuit consists of AC voltage source, rectifying element and resistive load connected in series. The circuit is shown in the figure (6.19). In half wave rectifier circuit there is a transformer at the input of the circuit .the transformer couples the AC voltage to the rectifier circuit . The function of the transformer is two fold . Firstly, it steps the source voltage up or down. Secondly, the transformer provides the isolate of ac power source from the rectifier circuit.

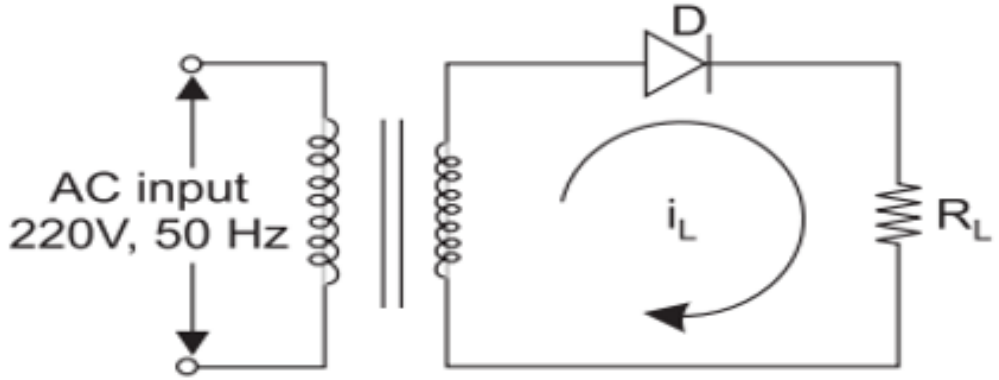


Figure (6.19): *Half-wave rectifier circuit*

Half Wave Rectifier

Operation of the circuit

During the positive half cycle of the input voltage, the diode is forward-biased. The current flows in the circuit in clockwise direction as shown in the figure 6.20(a). The current flowing through the load resistor R_L produces output voltage V_0 .The current flows for almost the positive half cycle. In forward bias, the voltage drop across the diode is very small. For practical purpose, we assume that diode is ideal so that the voltage drop across the diode is zero. Therefore, the output voltage V_0 is nearly same as the input voltage v_P at every instant.

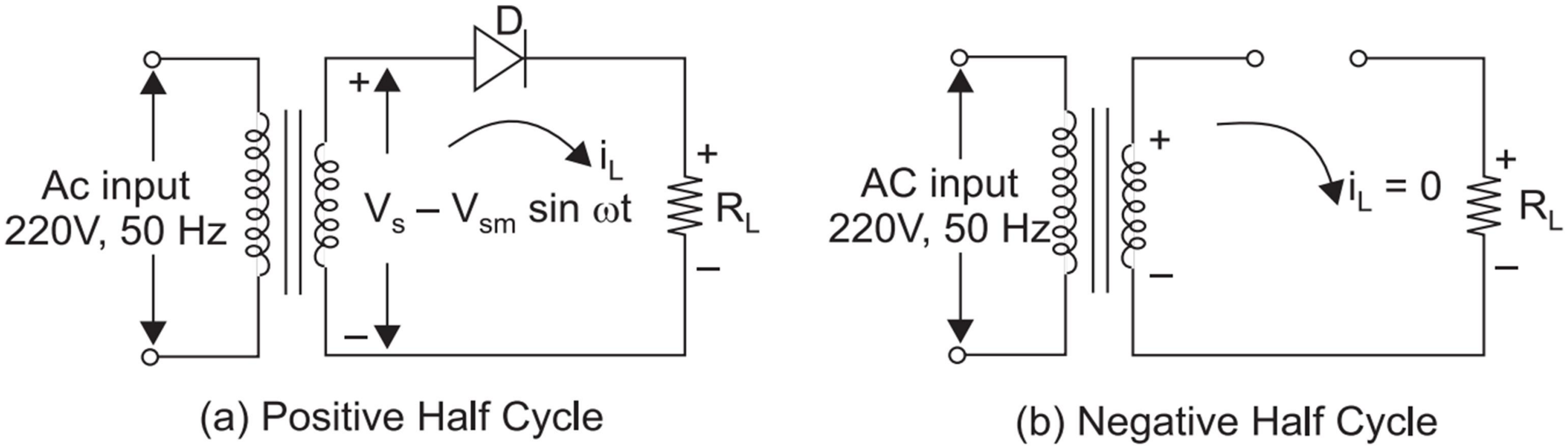


Figure (6.20): *Operation of half wave rectifier*

Half Wave Rectifier

Operation of the circuit

During the negative half cycle of the input ac voltage, the diode is reverse-biased shown in figure 6.20(b). Since, the diode is reverse-biased, it does not conduct .Therefore, current flows in the circuit ($i_L = 0$) and the output voltage across load resistor is also zero ($v_0 = 0$) Thus, from above it is that only the positive half cycle is utilized. During the positive half cycle, output voltage across the load is same as the input voltage. During negative cycle, no output voltage is available. This can be interpreted form the figure (6.21). Hence, the output voltage is not a perfect DC, it is unidirectional and pulsating voltage.

Half Wave Rectifier

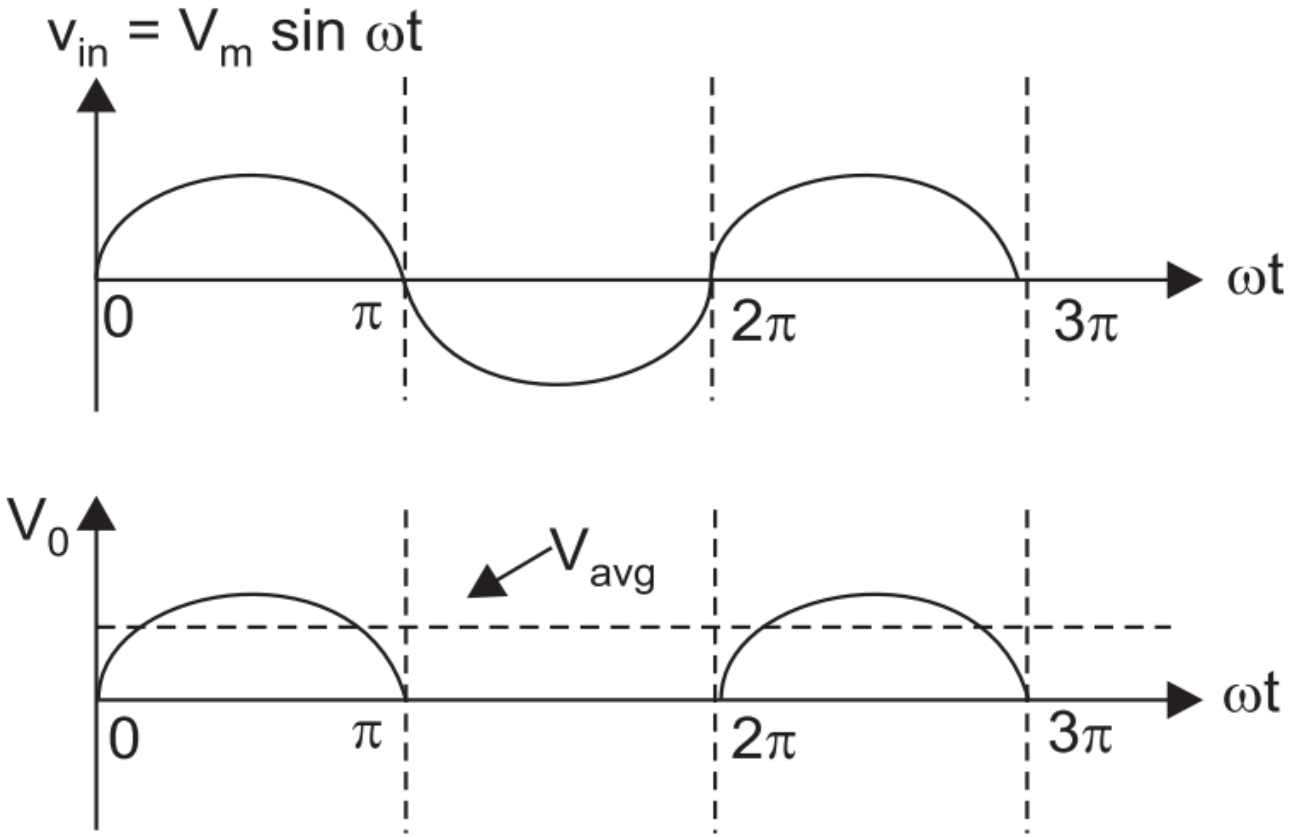


Figure (6.21): *Input and Output waveforms of half-wave rectifier.*

Ripple Factor (Γ)

- The ratio of rms value of a.c. component to the d.c. component in the output is known as ripple factor (Γ).

$$\Gamma = \frac{\text{rms value of a.c. component}}{\text{d.c. value of component}} = \frac{V_{r,\,\text{rms}}}{V_{\text{d.c.}}} \qquad \text{where} \quad V_{r,\,\text{rms}} = \sqrt{V_{\text{rms}}^2 - V_{\text{d.c.}}^2}$$

$$\Gamma = \sqrt{\left(\frac{V_{\text{rms}}}{V_{\text{d.c.}}}\right)^2 - 1}$$

- V_{av} is the average or the d.c. content of the voltage across the load and is given by

$$V_{av} = V_{\text{d.c.}} = \frac{1}{2\pi}\left[\int_0^{\pi} V_m \sin \omega t\, d(\omega t) + \int_{\pi}^{2\pi} 0.\text{d}(\omega t)\right]$$

$$= \frac{V_m}{2\pi}[-\cos \omega t]_0^{\pi} = \frac{V_m}{\pi}$$

$$I_{\text{d.c}} = \frac{V_{\text{d.c.}}}{R_L} = \frac{V_m}{\pi R_L} = \frac{I_m}{\pi}$$

- The rms voltage at the load resistance can be calculated as

$$V_{\text{rms}} = \left[\frac{1}{2\pi}\int_0^{\pi} V_m^2 \sin^2 \omega t \, \mathrm{d}(\omega t)\right]^{\frac{1}{2}}$$

$$= V_m \left[\frac{1}{4\pi}\int_0^{\pi} (1 - \cos 2\omega t)\, \mathrm{d}(\omega t)\right]^{\frac{1}{2}} = \frac{V_m}{2}$$

Therefore $$\Gamma = \sqrt{\left[\frac{V_m/2}{V_m/\pi}\right]^2 - 1} = \sqrt{\left(\frac{\pi}{2}\right)^2 - 1} = 1.21$$

- From this expression it is clear that the amount of a.c. present in the output is 121% of the d.c. voltage. So the half-wave rectifier is not practically useful in converting a.c. into d.c.

Efficiency (η)

- The ratio of d.c. output power to a.c. input power is known as rectifier efficiency (η).

$$\eta = \frac{\text{d.c. output power}}{\text{a.c. input power}} = \frac{P_{\text{d.c.}}}{P_{\text{a.c.}}}$$

$$= \frac{\dfrac{(V_{\text{d.c.}})^2}{R_L}}{\dfrac{(V_{\text{rms}})^2}{R_L}} = \frac{\left(\dfrac{V_m}{\pi}\right)^2}{\left(\dfrac{V_m}{2}\right)^2} = \frac{4}{\pi^2} = 0.406 = 40.6\%$$

- The maximum efficiency of a half-wave rectifier is 40.6%.

Peak Inverse Voltage (PIV)

- It is defined as the maximum reverse voltage that a diode can withstand without destroying the junction.
- The peak inverse voltage across a diode is the peak of the negative half cycle. For half-wave rectifier, PIV is V_m.

Form Factor (FF)

$$\text{Form factor} = \frac{\text{rms value}}{\text{average value}}$$

$$= \frac{V_m/2}{V_m/\pi} = \frac{\pi}{2} = 1.57$$

Peak Factor

$$\text{Peak factor} = \frac{\text{peak value}}{\text{rms value}}$$

$$= \frac{V_m}{V_m/2} = 2$$

Advantages:

- It is a simple circuit.
- Easy to construct.
- It has low number of components; therefore, it is cheap.

Disadvantages:

- The rectification efficiency of 40.6% indicates it is quite inefficient.
- The output contains much ripple content.

Full Wave Rectifier

- It converts an a.c. voltage into a pulsating d.c. voltage using both half cycles of the applied a.c. voltage. It uses two diodes of which one conducts during one half-cycle while the other diode conducts during the other half-cycle of the applied a.c. voltage.
- There are two types of full-wave rectifiers viz. (i) Full-wave rectifier with center tapped transformer and (ii) Full-wave rectifier without transformer (Bridge rectifier).
- Fig. 4 shows the basic circuit and waveforms of full-wave rectifier. During positive half of the input signal, anode of diode D_1 becomes positive and at the same time the anode of diode D_2 becomes negative. Hence, D_1 conducts and D_2 does not conduct. The load current flows through D_1 and the voltage drop across R_L will be equal to the input voltage.

Full Wave Rectifier

- During the negative half-cycle of the input, the anode of D_1 becomes negative and the anode of D_2 becomes positive. Hence, D_1 does not conduct and D_2 conducts. The load current flows through D_2 and the voltage drop across R_L will be equal to the input voltage.

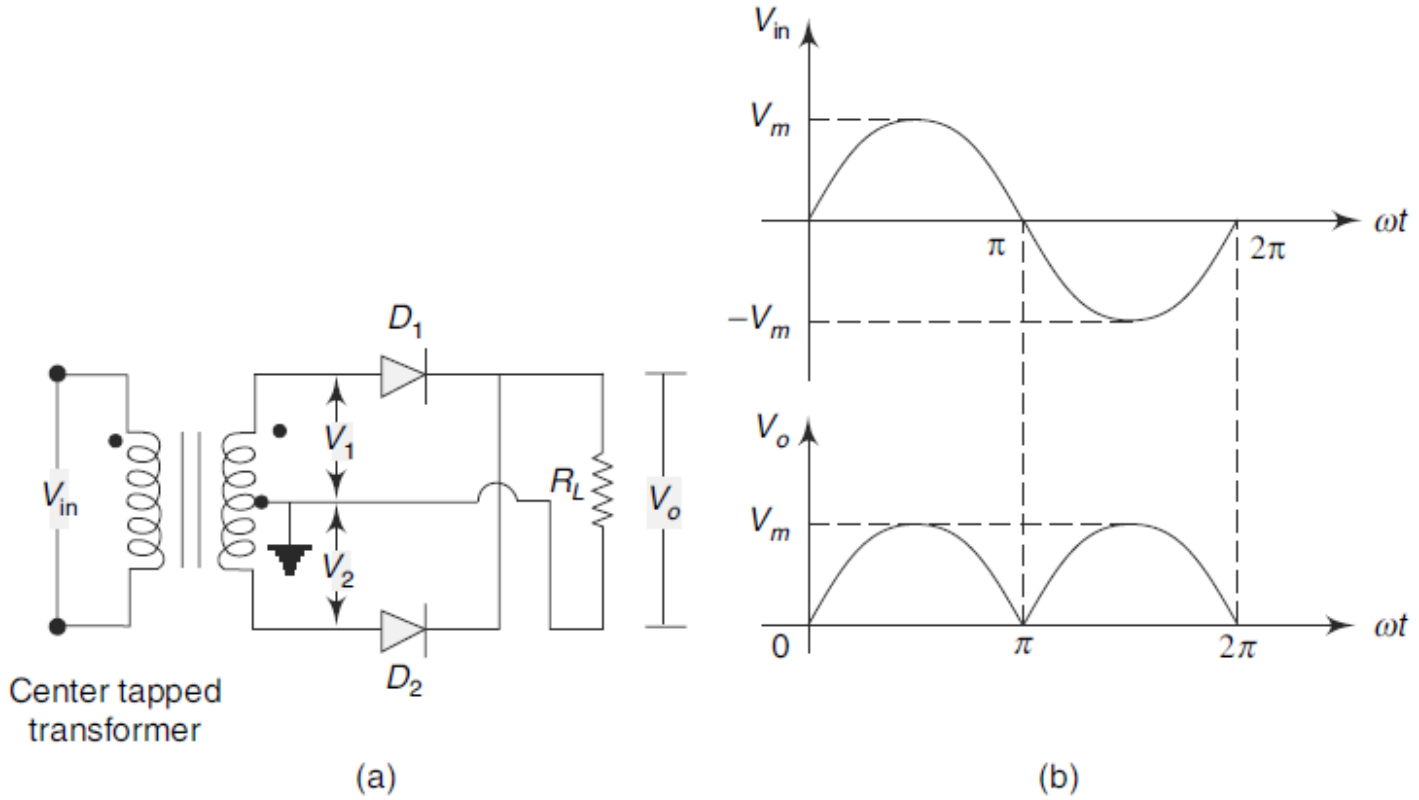


Fig. 4 Full wave rectifier (a)basic circuit and (b)waveforms.

Full Wave Rectifier

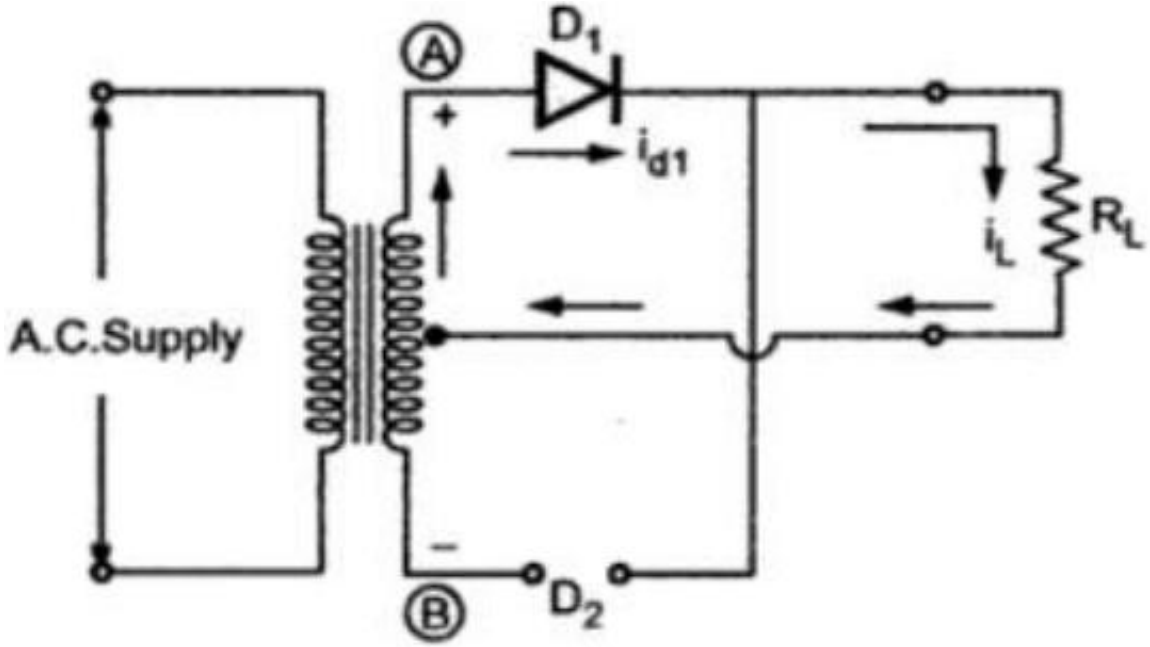


Fig. Current flow during positive half cycle

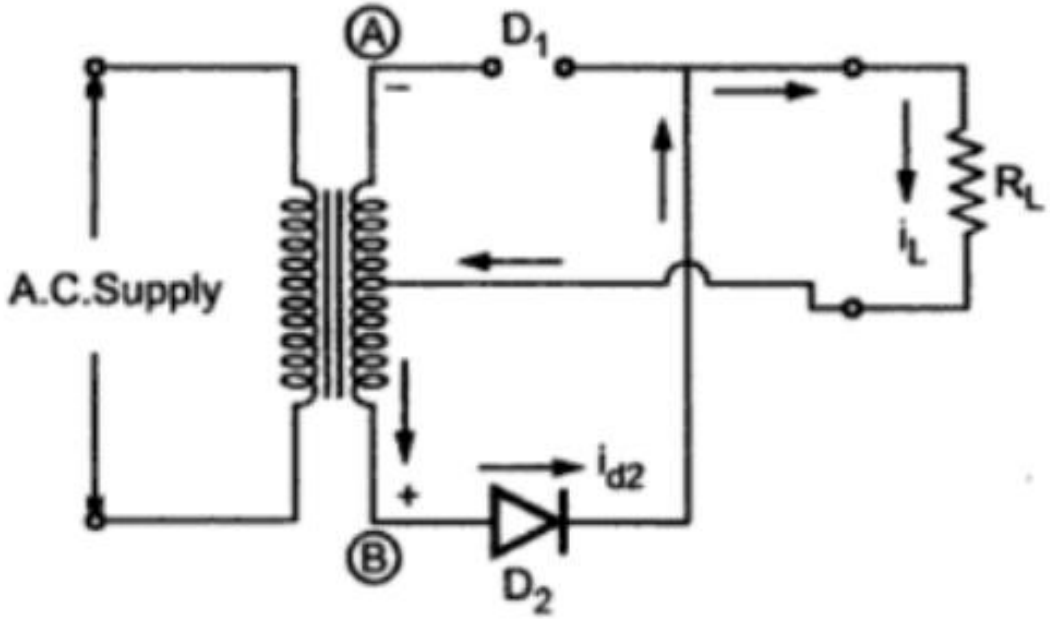


Fig. Current flow during negative half cycle

Ripple Factor (Γ)

$$\Gamma = \sqrt{\left(\frac{V_{\text{rms}}}{V_{\text{d.c.}}}\right)^2 - 1}$$

- The average voltage or d.c. voltage available across the load resistance is

$$V_{\text{d.c.}} = \frac{1}{\pi}\int_0^{\pi} V_m \sin \omega t \, \mathrm{d}(\omega t)$$

$$= \frac{V_m}{\pi}[-\cos \omega t]_0^{\pi} = \frac{2V_m}{\pi}$$

$$I_{\text{d.c.}} = \frac{V_{\text{d.c.}}}{R_L} = \frac{2V_m}{\pi R_L} = \frac{2I_m}{\pi} \text{ and } I_{\text{rms}} = \frac{I_m}{\sqrt{2}}$$

- RMS value of the voltage at the load resistance is

$$V_{\text{rms}} = \sqrt{\left[\frac{1}{\pi}\int_0^{\pi} V_m^2 \sin^2 \omega t \, \mathrm{d}(\omega t)\right]} = \frac{V_m}{\sqrt{2}}$$

Therefore,

$$\Gamma = \sqrt{\left(\frac{V_m/\sqrt{2}}{2V_m/\pi}\right)^2 - 1} = \sqrt{\frac{\pi^2}{8} - 1} = 0.482$$

Efficiency (η)

- The ratio of d.c. output power to a.c. input power is known as rectifier efficiency (η).

$$\eta = \frac{\text{d.c. output power}}{\text{a.c. input power}} = \frac{P_{\text{d.c.}}}{P_{\text{a.c.}}}$$

$$= \frac{(V_{\text{d.c.}})^2/R_L}{(V_{\text{rms}})^2/R_L} = \frac{\left[\frac{2V_m}{\pi}\right]^2}{\left[\frac{V_m}{\sqrt{2}}\right]^2} = \frac{8}{\pi^2} = 0.812 = 81.2\%$$

- The maximum efficiency of a full-wave rectifier is 81.2%.

Transformer Utilization Factor (TUF)

- The average TUF in full-wave rectifying circuit is determined by considering the primary and secondary windings separately and it gives a value of 0.693.

Form Factor (FF)

$$\text{Form factor} = \frac{\text{rms value of the output voltage}}{\text{average value of the output voltage}}$$

$$= \frac{V_m/\sqrt{2}}{2V_m/\pi} = \frac{\pi}{2\sqrt{2}} = 1.11$$

Peak Factor

$$\text{Peak factor} = \frac{\text{peak value of the output voltage}}{\text{rms value of the output voltage}} = \frac{V_m}{V_m/\sqrt{2}} = \sqrt{2}$$

- Peak inverse voltage for full-wave rectifier is $2V_m$ because the entire secondary voltage appears across the non-conducting diode.

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Advantages of full-wave rectifier

- The ripple frequency is two times the input frequency.
- Efficiency is higher.
- The large DC power output
- Ripple factor is less
- Higher output voltage
- Higher transformer utilization factor
- Utilizes both halves of the AC waveform

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Disadvantages of full-wave rectifier

- More complicated than half-wave rectifier
- It requires more diodes
- PIV rating of the diode is higher
- Higher PIV diodes are larger in size and too much costlier
- The cost of the centre tap transformer is high

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Bridge Rectifier

- The need for a center tapped transformer in a full-wave rectifier is eliminated in the bridge rectifier. As shown in Fig. 5, the bridge rectifier has four diodes connected to form a bridge.
- The a.c. input voltage is applied to the diagonally opposite ends of the bridge. The load resistance is connected between the other two ends of the bridge.
- For the positive half-cycle of the input a.c. voltage, diodes D1 and D3 conduct, whereas diodes D2 and D4 do not conduct. The conducting diodes will be in series through the load resistance RL. So the load current flows through RL.
- During the negative half-cycle of the input a.c. voltage, diodes D2 and D4 conduct, whereas diodes D1 and D3 do not conduct. The conducting diode D2 and D4 will be in series through the load RL and the current flows through RL in the same direction as in the previous half-cycle. Thus a bidirectional wave is converted into an unidirectional one.

Bridge Rectifier

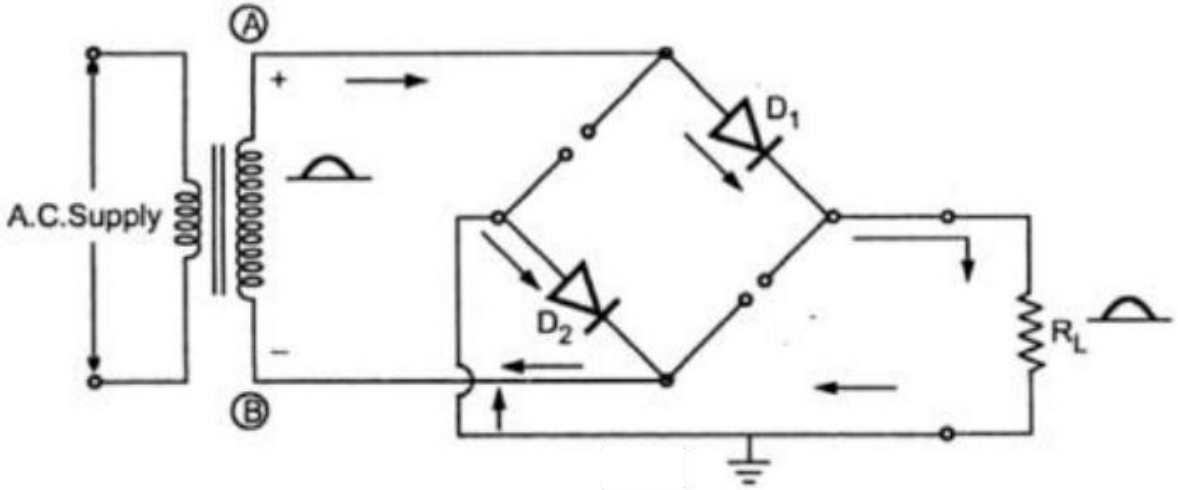


Fig. Current flow during positive half cycle

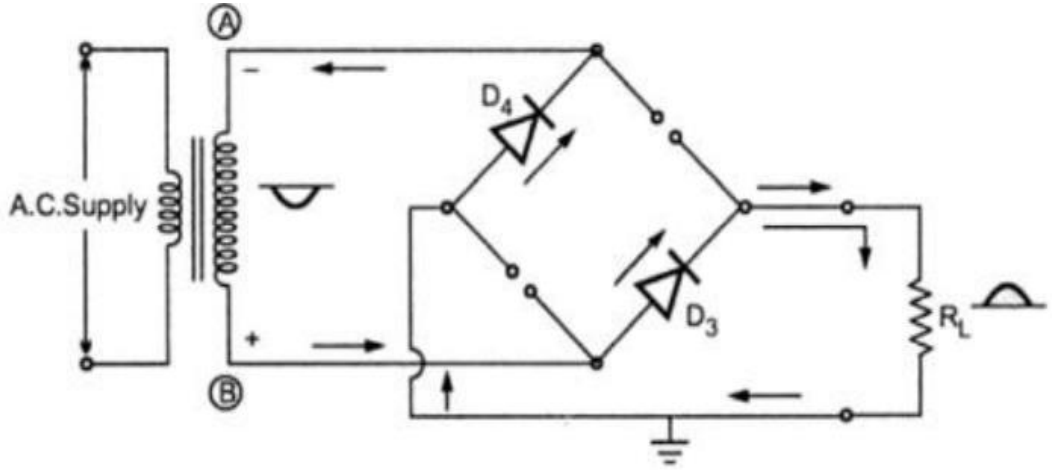
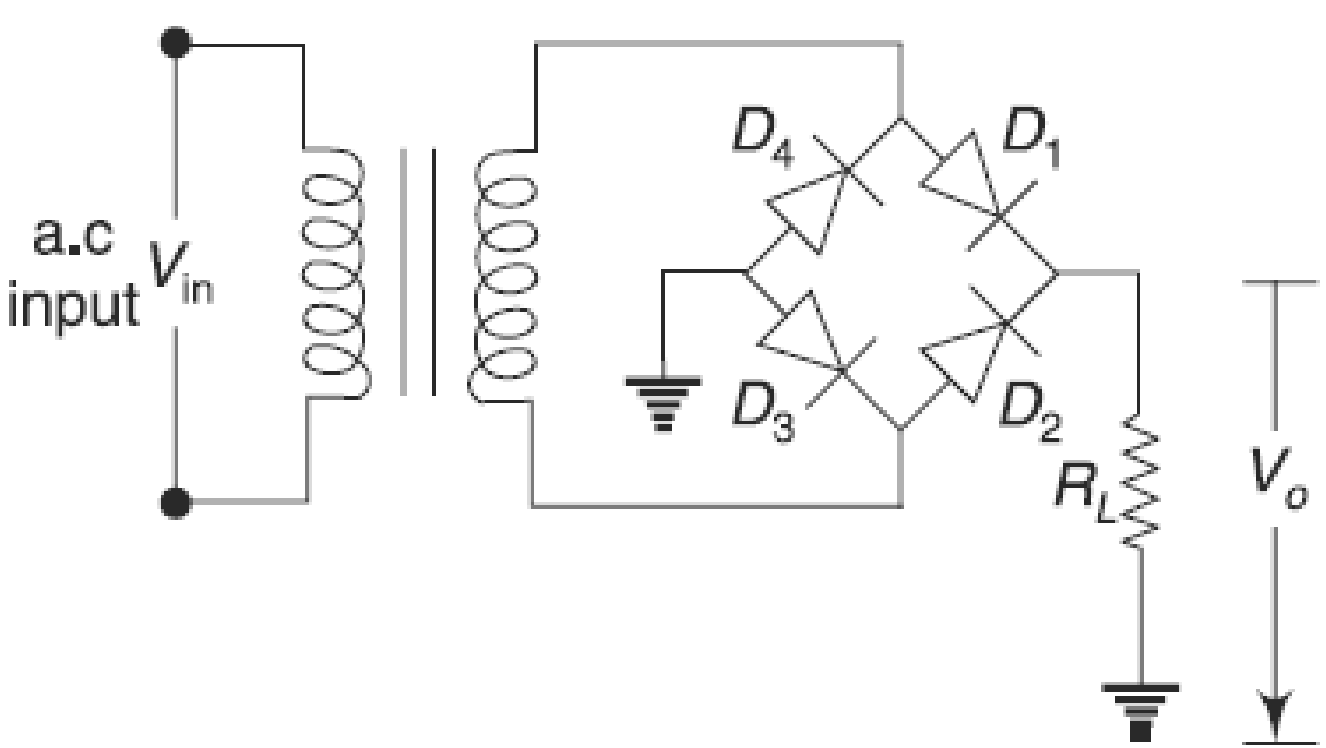
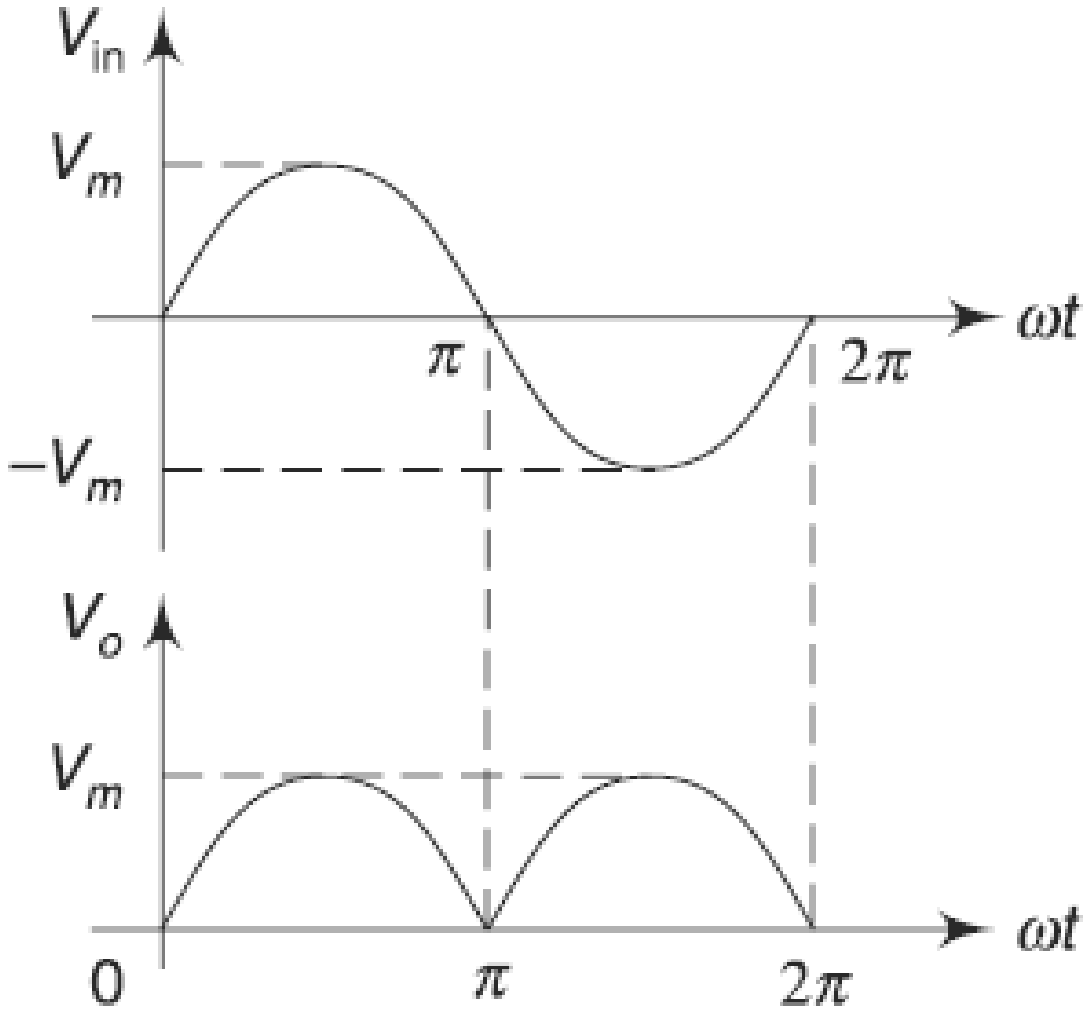


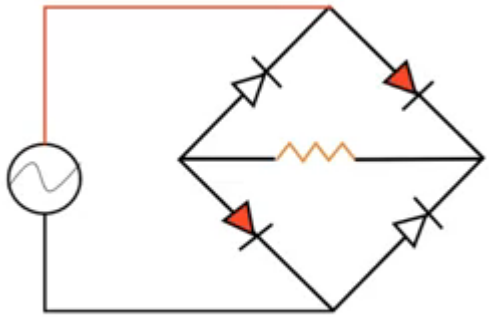
Fig. Current flow during negative half cycle

Bridge Rectifier





Bridge Rectifier



- The average values of output voltage and load current for bridge rectifier are the same as for a center tapp full-wave rectifier. Hence,

$$V_{\text{d.c.}} = \frac{2V_m}{\pi} \quad \text{and} \quad I_{\text{d.c.}} = \frac{V_{\text{d.c.}}}{R_L} = \frac{2V_m}{\pi R_L} = \frac{2I_m}{\pi}$$

- The maximum efficiency of a bridge rectifier is 81.2% and the ripple factor is 0.48. The PIV is V_m.

Comparison of rectifiers

Particulars	*Type of rectifier*		
	Half-wave	*Full-wave*	*Bridge*
No. of diodes	1	2	4
Maximum efficiency	40.6%	81.2%	81.2%
$V_{d.c.}$ (no load)	V_m/π	$2V_m/\pi$	$2V_m/\pi$
Average current/diode	$I_{d.c.}$	$I_{d.c.}/2$	$I_{d.c.}/2$
Ripple factor	1.21	0.48	0.48
Peak inverse voltage	V_m	$2V_m$	V_m
Output frequency	f	$2f$	$2f$
Transformer utilisation factor	0.287	0.693	0.812
Form factor	1.57	1.11	1.11
Peak factor	2	$\sqrt{2}$	$\sqrt{2}$

Filter

❖The output of a rectifier contains d.c. component as well as a.c. component. Filters are used to minimise the undesirable a.c., i.e. ripple leaving only the d.c. component to appear at the output. The ripple in the rectified wave being very high, the factor being 48% in the full wave rectifier; majority of the applications which cannot tolerate this, will need an output which has been further processed. Figure 3.8 shows the concept of a filter, where the full-wave rectified output voltage is applied at its input. The output of a filter is not exactly a constant d.c. level. But it also contains a small amount of a.c. component. Some important filters are:

(i) Inductor filter (ii) Capacitor filter (iii) LC or L-section filter (iv) CLC or p-type filter

Filter

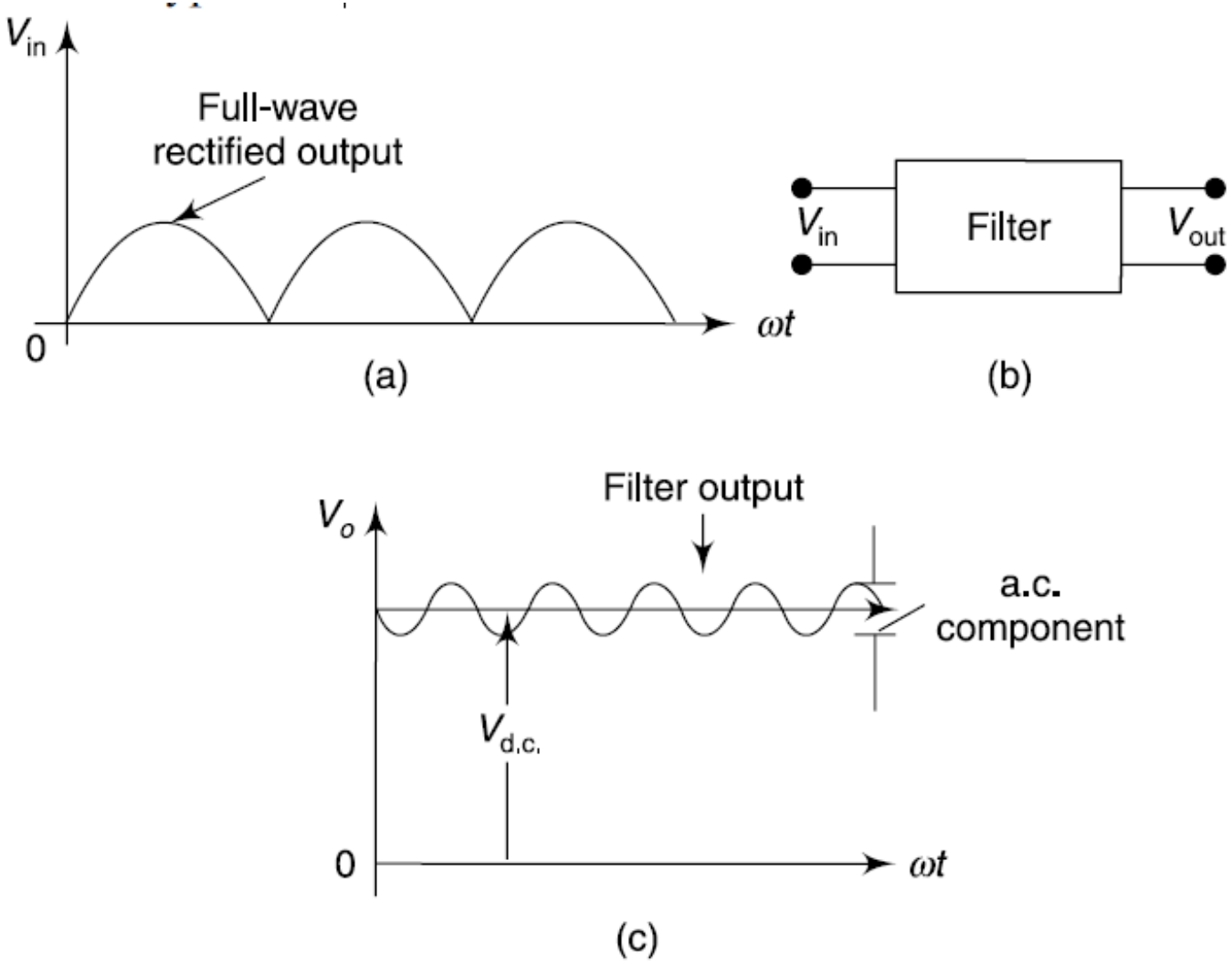


Fig. 3.8 *Concept of a filter*

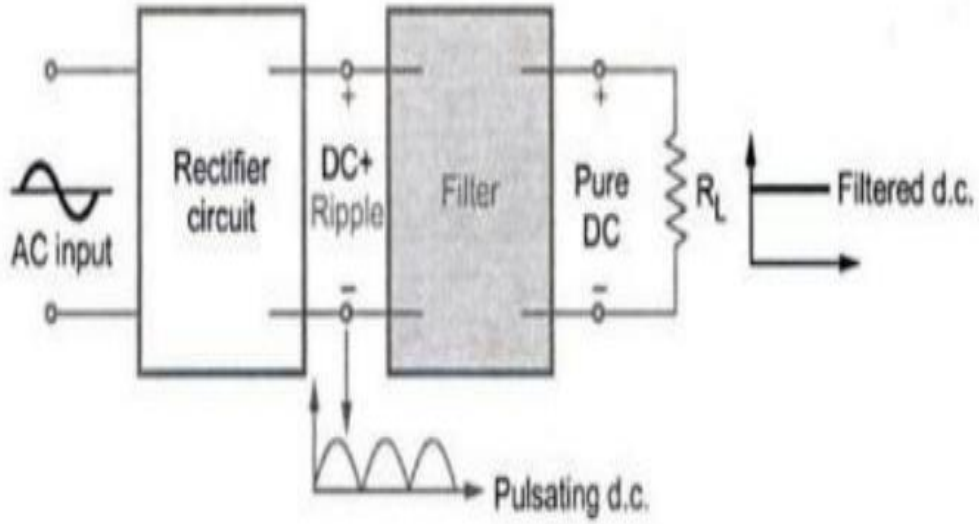
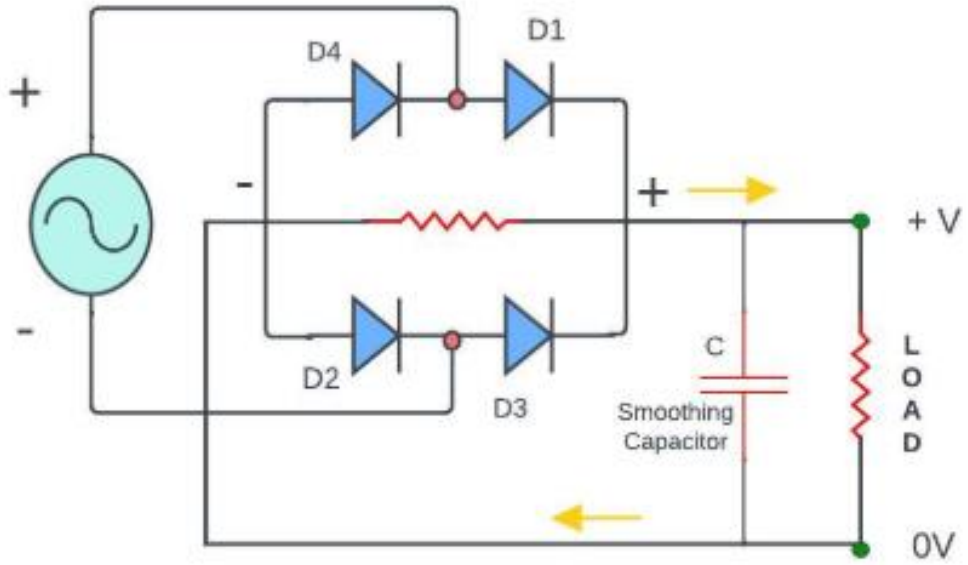


Fig. Power supply using rectifier and filter

Capacitor Input Filter or
Capacitor filter or
Full Wave Rectifier with Capacitor Filter

In a full-wave rectifier both positive and negative halves of the AC input waveform are used. This type of filter with capacitor gives more efficiency for converting AC to DC compared to a half-wave rectifier because it utilizes both halves of AC waveform that resulting in a smoother DC output with reduced ripple.



full Wave Bridge Rectifier with Capacitor Filter

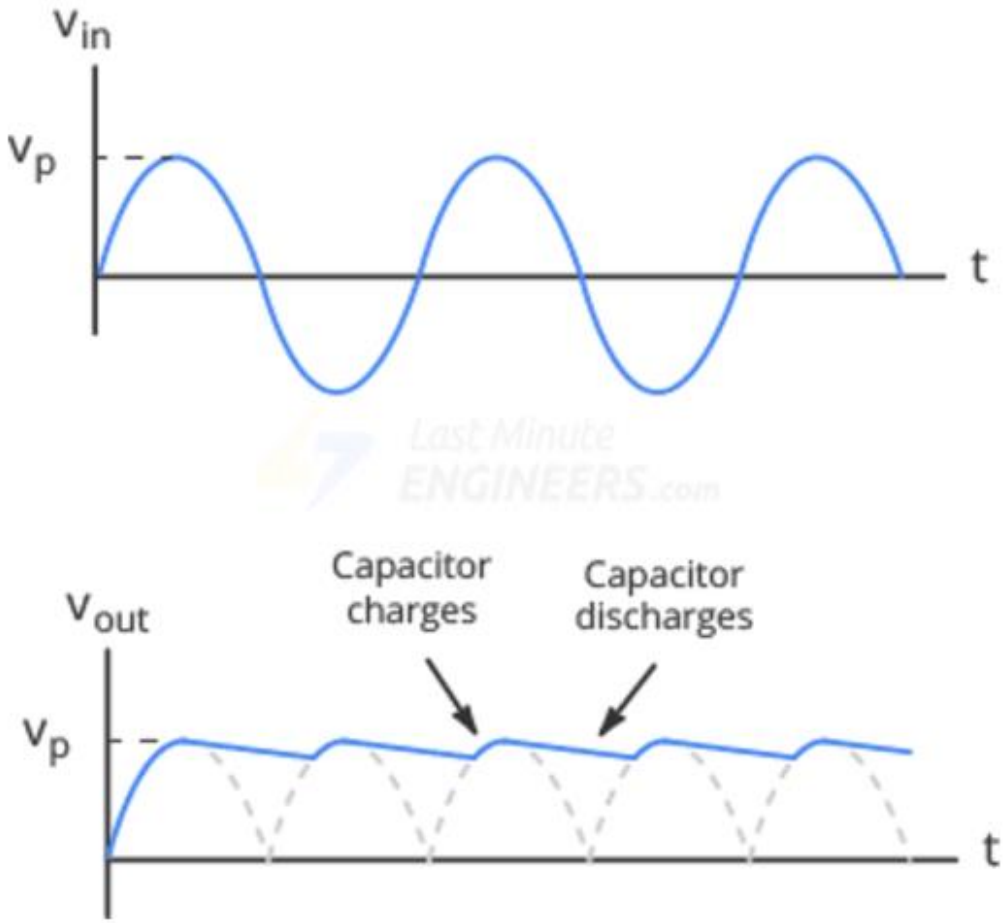
Capacitor Input Filter or
Capacitor filter or
Full Wave Rectifier with Capacitor Filter

This is typically Bridge rectifier configuration which utilizes four arranged in a bridge-like pattern. diodes are ensured that the current flows in the same direction through the load during both half (positive and negative) AC cycle resulting in a more continuous DC output.

Case I: during the positive half-cycle of the AC input one pair of diodes (D1 & D2) are conducts that allowing the capacitor to charge to peak value of the rectified voltage.

Case II: during the negative half-cycle the other one diode pair (D3 & D4) are conducts that allowing the capacitor to discharge into the load.

Capacitor Input Filter or
Capacitor filter or
Full Wave Rectifier with Capacitor Filter



full Wave Bridge Rectifier with Capacitor Filter waveform

Capacitor Input Filter or
Capacitor filter or
Full Wave Rectifier with Capacitor Filter

Output voltage:

The voltage across the capacitor helps to smooth out of the pulsating DC obtained from the full-wave rectifier.

Here is the output voltage is the sum of the DC components and reduced ripple voltage due to the capacitor filter.

Ripple Voltage:

The ripple voltage in full-wave rectifier with a capacitor is generally lower than half-wave rectifier with a similar filter.

Capacitor Input Filter or
Capacitor filter or
Full Wave Rectifier with Capacitor Filter

The advantages of capacitor input filter are,

1. Less number of components.
2. Low ripple factor hence low ripple voltage.
3. Suitable for high voltage at small load currents.

The disadvantages of capacitor input filter are,

1. Ripple factor depends on load resistance
2. Regulation is poor
3. Diodes are subjected to high surge currents hence must be selected accordingly.

Inductor Filter

In this type of filter, an inductor (choke) is connected in series with the load. It is known that the inductor opposes change in the current.

So the ripple which is change in the current is opposed by the inductor and it tries to smoothen the output .

Consider the full wave rectifier with inductor filter which is also called choke filter. Fig. 1.23 (a) shows the circuit diagram while Fig. 1.23(b) shows the current waveform obtained by using choke filter with full wave rectifier

Inductor Filter

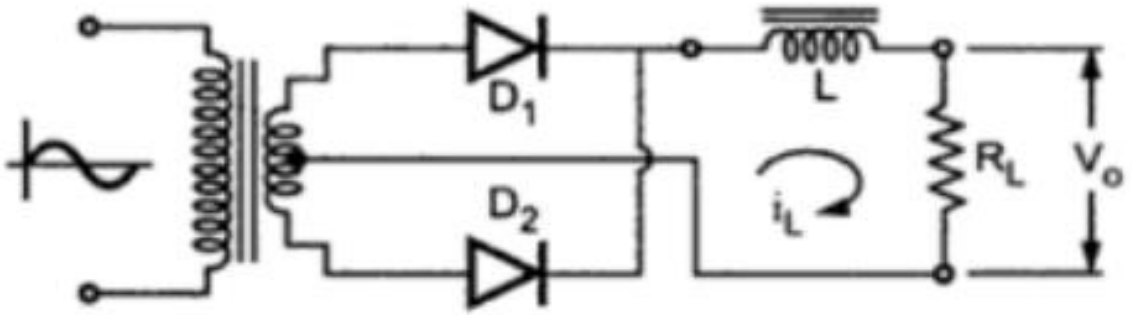


Fig.1.23 (a) Circuit diagram of choke filter

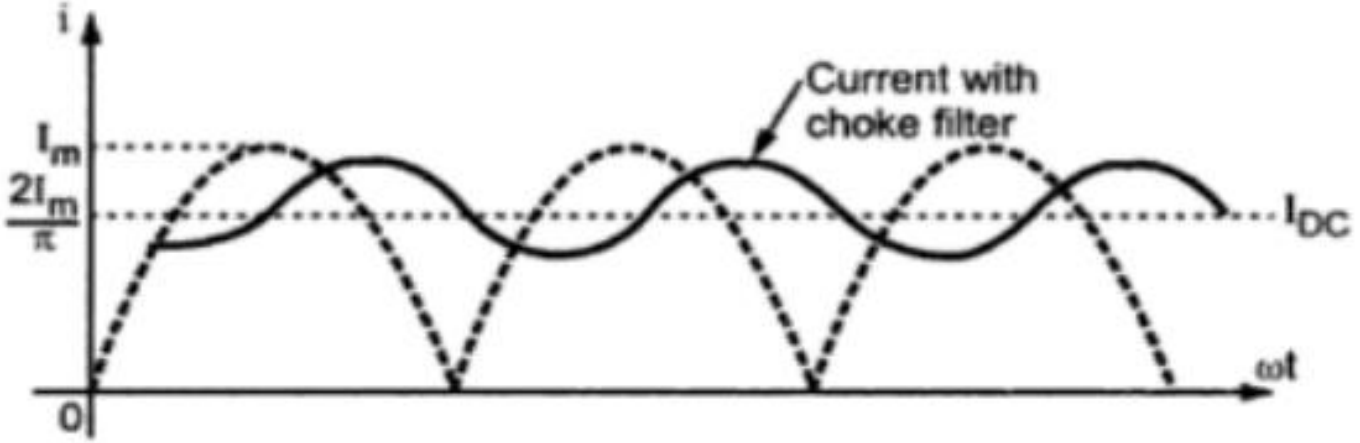


Fig.1.23 (a) Current waveform of choke filter

Inductor Filter

In the positive half cycle of the secondary voltage of the transformer, the diode D_1 is forward biased. Hence the current flows through D_1 ,L and R_L While in the negative half cycle, the diode D_1 is reverse biased while in the negative half cycle, the diode D_1 is reverse biased while diode $D_2$is forward biased. Hence the current flows through D_2 ,L and R_L. Hence we get unidirectional current through R_L. Due to inductor L which opposes change in current, it tries to make the output smooth by opposing the ripple content in the output.

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INTRODUCTION

- Digital electronics find applications in computers, calculators, integrated circuits, and many digital devices and circuits. Digital circuits are being used in many applications like in display devices, digital clocks, digital counters, digital instruments, etc.
- Digital electronics includes electronic systems that use digital signals. Digital electronics or any digital circuit are usually made from large assemblies of logic gates. Digital electronics, digital circuits, and digital systems are often used interchangeably.
- Digital circuits and systems used in computers and electronic controls are logic gates and multivibrators.
- Logic gates are combinational (i.e., with no memory) and sequential (i.e., with memory).
- Combinational gates are AND, NOT, OR, NAND and NOR gates. Sequential gates include flip-fl ops (bistable multivibrators) which are available as registers and counters

Overview of Number Systems

A number system is a set of rules of symbols used to represent numbers. There are number systems like decimal, binary, octal, and hexadecimal. The knowledge of number systems is very important for the design of digital circuits and systems.

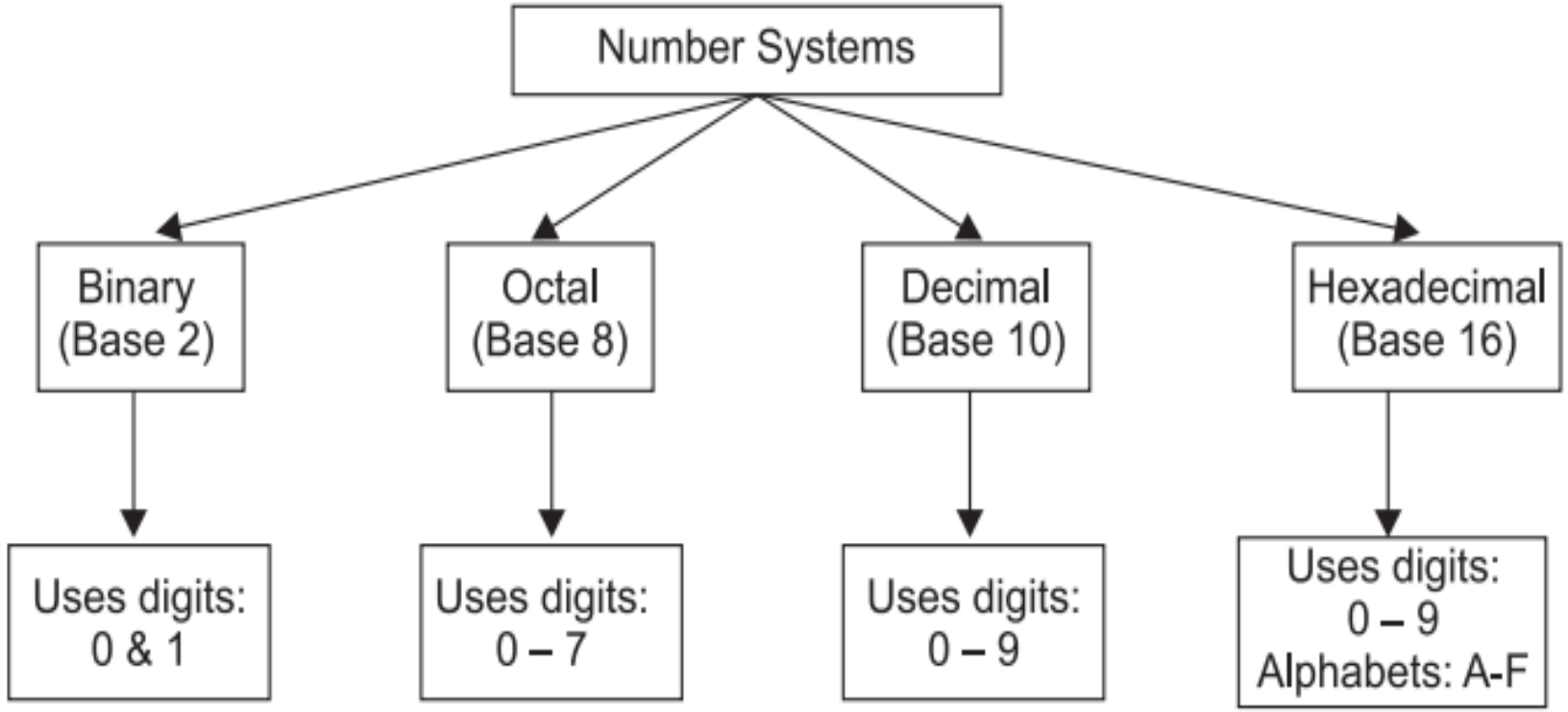


Figure (8.1): *Types of number systems*

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Decimal (Base 10) Number system:

The decimal number system has base 10 as it uses 10 digits from 0 to 9, i.e., 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. This system uses positional notation, i.e., the least-significant digit (right-most digit) is of the order of 10° (units or ones), the second right-most digit is of the order of 10^1 (tens), the third right-most digit is of the order of 10^2 (hundreds), and so on.

10^5	10^4	10^3	10^2	10^1	10^0

For example,

$$6432 = 6 \times 10^3 + 4 \times 10^2 + 3 \times 10^1 + 2 \times 10^0 = 6000 + 400 + 30 + 2$$

For example, 12265 has place values as,

$(1 \times 10^4) + (2 \times 10^3) + (2 \times 10^2) + (6 \times 10^1) + (5 \times 10^0)$

$= (1 \times 10000) + (2 \times 1000) + (2 \times 100) + (6 \times 10) + (5 \times 1)$

$= 10000 + 2000 + 200 + 60 + 5$

$= 12265$

Binary (Base 2) Number system:

The binary number system uses only two basic digits such as 0 and 1. Each digit in a binary number is called a bit. The right most bit of a binary number is called as least significant bit (LSB) whereas the left most bit of a binary number is called as most significant bit (MSB). The LSB has a weight or value of 2^0, the second from the right has a value of 2^1, the next 2^2 etc. In case of fractions, the weights starting from the right of the fraction point are 2^{-1}, 2^{-2}, 2^{-3} and so on.

2^5	2^4	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}	2^{-3}	2^{-4}	2^{-5}

Any number can be expressed in binary form in the usual way. Table (8.1) shows representation of binary numbers.

Binary (Base 2) Number system:

Table (8.1): *Counting binary numbers*

2^3	2^2	2^1	2^0	Binary Number	Decimal Number
0	0	0	0	0000	0
0	0	0	1	0001	1
0	0	1	0	0010	2
0	0	1	1	0011	3
0	1	0	0	0100	4
0	1	0	1	0101	5
0	1	1	0	0110	**6**
0	1	1	1	0111	**7**
1	0	0	0	1000	**8**
1	0	0	1	1001	9
1	0	1	0	1010	10
1	0	1	1	1011	11
1	1	0	0	1100	12
1	1	0	1	1101	13
1	1	1	0	1110	14
1	1	1	1	1111	15

Binary (Base 2) Number system:

A binary number can be converted into an equivalent decimal number.

Solved Problem-1: Convert binary number (101101) into decimal number.

Solution:

$$(101101)_2 = 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 1 \times 32 + 0 \times 16 + 1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1$$

$$= 32 + 8 + 4 + 1$$

$$= (45)_{10}$$

Therefore, $(101101)_2 = (45)_{10}$

Binary (Base 2) Number system:

In the binary system a number is expressed by two symbols or digits, 0 and 1. Since the total number of symbols or digits used is two, it has a base 2. The binary symbols 0 and 1 are called binary digits or bits. A group of eight bits is called a byte. In this system the positional value is used in the same way as the decimal system. For example, two binary numbers, 11010 and 110011 are represented as their decimal equivalent as

Number	Value	Position	Positional value	Decimal equivalent
1 1 0 1 0				
	0	0	2^0	$0 \times 2^0 = 0$
	1	1	2^1	$1 \times 2^1 = 2$
	0	2	2^2	$0 \times 2^2 = 0$
	1	3	2^3	$1 \times 2^3 = 8$
	1	4	2^4	$1 \times 2^4 = 16$
				26
1 1 0 0 1 1				
	1	0	2^0	$1 \times 2^0 = 1$
	1	1	2^1	$1 \times 2^1 = 2$
	0	2	2^2	$0 \times 2^2 = 0$
	0	3	2^3	$1 \times 2^3 = 0$
	1	4	2^4	$1 \times 2^4 = 16$
	1	5	2^5	$1 \times 2^5 = 32$
				51

Octal(Base 8) Number system:

The octal number system has base 8 as it uses 8 digits from 0 to 7, i.e., 0, 1, 2, 3, 4, 5, 6, 7. Each digit in the octal number system corresponds to the powers of 8. The weight of positions in .....4, 3, 2, 1, 0, –1, – 2, – 3 ..... octal number is as follows: 8^4 8^3 8^2 8^1 8^0. 8^{-1} 8^{-2} 8^{-3} 8^{-4}... as given below:

8^5	8^4	8^3	8^2	8^1	8^0	8^{-1}	8^{-2}	8^{-3}	8^{-4}	8^{-5}

An octal number can be converted into an equivalent decimal number.

Solved Problem-2: Convert octal number $(12570)_8$ into decimal number.

Solution:

$$(12570)_8 = 1 \times 8^4 + 2 \times 8^3 + 5 \times 8^2 + 7 \times 8^1 + 0 \times 8^0$$
$$= 1 \times 4096 + 2 \times 512 + 5 \times 64 + 7 \times 8 + 0 \times 1$$
$$= 4096 + 1024 + 320 + 56 + 0$$
$$= (5496)_{10}$$

Therefore, $(12570)_8 = (5496)_{10}$

Hexadecimal (Base 16) Number system:

The hexadecimal number system has ten digits from 0 to 9 and six alphabets A to F. The weight of each digit of hexadecimal number is ….16^4 16^3 16^2 16^1 16^0. 16^{-1} 16^{-2} 16^{-3} 16^{-4}…. as given below:

16^5	16^4	16^3	16^2	16^1	16^0	16^{-1}	16^{-2}	16^{-3}	16^{-4}	16^{-5}

Table (8.2) gives the equivalent binary representation of decimal and hexadecimal number systems.

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Hexadecimal (Base 16) Number system:

Table (8.2): *Equivalent binary representation of decimal and hexadecimal number system*

Decimal Number	Hexadecimal Number	Binary Number
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111

Hexadecimal (Base 16) Number system:

Solved Problem-3: Convert given hexadecimal number $(19FDE)_{16}$ into its equivalent decimal number.

Solution: $(19FDE)_{16} = 1 \times 16^4 + 9 \times 16^3 + F \times 16^2 + D \times 16^1 + E \times 16^0$

$= 1 \times 16^4 + 9 \times 16^3 + 15 \times 16^2 + 13 \times 16^1 + 14 \times 16^0$

$= 65536 + 36864 + 3840 + 208 + 14$

$= (106462)_{10}$

Therefore, $(19FDE)_{16} = (106462)_{10}$

Hexadecimal (Base 16) Number system:

The hexadecimal equivalent of a decimal number can be obtained by dividing the decimal number by 16 repeatedly, until a quotient of 0 is obtained. The following example illustrates how the hexadecimal equivalent of a given decimal can be obtained.

Example Convert 112_{10} and 253_{10} to hexadecimal numbers.

The procedure is as follows,

Solution: (i) 112 divided by 16 = quotient 7 with a remainder of **0**

7 divided by 16 = quotient 0 with a remainder of **7**

Reading the remainders from bottom to top, the decimal number 112_{10} is equivalent to hex 70_{16}. The hexadecimal 70_{16} can also be represented as 70 H.

(ii) 253 divided by 16 = quotient 15 with a remainder of **13**, i.e *D*

15 divided by 16 = quotient 0 with a remainder of **15**, i.e. *F*

Reading the remainders from bottom to top, the decimal number 253_{10} is equivalent to hex FD_{16}.

Hexadecimal (Base 16) Number system:

Example Convert the following hexadecimal numbers to decimals.

(i) $4AB$ H (ii) $23F$ H

Solution: (i) $4AB_{16} = 4 \times 16^2 + A \times 16^1 + B \times 16^0$

$= 4 \times 16^2 + 10 \times 16^1 + 11 \times 16^0$

$= 4 \times 256 + 10 \times 16 + 11 \times 1$

$= 1024 + 160 + 11$

$= 1195_{10}$

(ii) $23F_{16} = 2 \times 16^2 + 3 \times 16^1 + F \times 16^0$

$= 2 \times 256 + 3 \times 16 + 15 \times 1$

$= 512 + 48 + 15$

$= 575_{10}$

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Binary Coded Decimal (BCD) code :

Binary Coded Decimal or BCD, is another process for converting decimal numbers into their binary equivalents. In this code each decimal digit is represented by a 4-bit binary number. In the binary system, with four bits we can represent sixteen numbers (0000 to 1111). But in BCD code only first ten of these are used (0000 to 1001). The remaining six code combinations i.e., 1010 to 1111 are invalid in BCD.

In the BCD numbering system, the given decimal number is segregated into chunks of four bits for each decimal digit within the number. Each decimal digit is converted into its direct binary form. Table (8.3) gives the binary and BCD codes for the decimal numbers 0 to 15.

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Binary Coded Decimal (BCD) code :

Table (8.3): *BCD code representation*

Decimal Number	Binary Number	Binary Coded Decimal
0	0000	0000
1	0001	0001
2	0010	0010
3	0011	0011
4	0100	0100
5	0101	0101
6	0110	0110
7	0111	0111
8	1000	1000
9	1001	1001
10	1010	0001 0000
11	1011	0001 0001
12	1100	0001 0010
13	1101	0001 0011
14	1110	0001 0100
15	1111	0001 0101

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Binary Coded Decimal (BCD) code :

Advantages:

- It is very similar to decimal system.
- We need to remember binary equivalent of decimal numbers 0 to 9 only.

Disadvantages:

- The addition and subtraction of BCD have different rules.
- The BCD arithmetic is little more complicated.
- BCD needs more number of bits than binary to represent the decimal number. So, BCD is less efficient than binary.

Decimal-to-BCD Conversion

The conversion of decimal to binary coded decimal is very similar to the conversion of hexadecimal to binary. Firstly, separate the decimal number into its weighted digits and then write down the equivalent 4-bit BCD code representing each decimal digit.

Examples:

$$85_{10} = (1000 \quad 0101)_{BCD}$$

$$572.73_{10} = (0101 \quad 0111 \quad 0010. \quad 0111 \quad 0011)_{BCD}$$

BCD-to- Decimal Conversion:

The conversion from binary coded decimal to decimal is the exact opposite of the above. Simply divide the binary number into groups of four digits, starting with the least significant digit and then write the decimal digit represented by each 4-bit group. Add additional zero's at the end if required to produce a complete 4-bit grouping. So, for example, 110101_2 would become: $(0011\ 0101)_2$ or $(35)_{10}$ in decimal.

Examples:

1001 in BCD represents the decimal digit 9.

$1000111_2 = 0100\ 0111_{BCD} = 47_{10}$

Excess-3 CODE:

Excess-3 code, also known as XS-3 code, is a non-weighted and self-complementary BCD code used to represent decimal numbers. It adds a fixed value of 3 to each decimal digit before converting it into a binary code or it can also add 0011 in each 4-bit BCD code of the decimal number for getting excess-3 code. The Excess-3 code for the decimal numbers are shown in Table 8.4.

We can find the excess-3 code of the given binary number by using the following steps:

1. Convert the binary number into decimal.
2. Add 3 in each digit of the decimal number.
3. Find the binary code of each digit of the newly generated decimal number.

We can also add 0011 in each 4-bit BCD code of the decimal number for getting excess-3 code.

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Excess-3 CODE:

Table (8.4): *Excess-3 code for the decimal numbers*

Decimal Code	BCD Code	Excess-3 Code
0	0000	0011
1	0001	0100
2	0010	0101
3	0011	0110
4	0100	0111
5	0101	1000
6	0110	1001
7	0111	1010
8	1000	1011
9	1001	1100

Excess-3 CODE:

- The codes 0000 and 1111 are not used for any digit which is an advantage for memory organization as these codes can cause fault in transmission line.
- It has no limitation, and it considerably simplifies arithmetic operations.

Example:

(i) $(32)_{10} = (0011\ 0010)_{BCD} = (0110\ 0101)_{excess\text{-}3}$

(ii) $(72.51)_{10} = (0111\ 0010.0101\ 0001)_{BCD} = (1010\ 0101.1000\ 0100)_{excess\text{-}3}$

Excess-3 code is self-complementary binary code means which is always complimented in itself. By replacing the bit 0 to 1 and 1 to 0 of a number is called 1's complement of the number. The 1's complement of excess-3 of a decimal number, it will be equal to the excess-3 code of the 9's complement of that decimal number.

For example: The excess-3 code of decimal 5 is 1000, the complement value will be 0111, which is the excess-3 code for decimal 4 which is 9's complement of 5.

Excess-3 CODE:

Solved Problem-4: Convert the following decimal numbers into Excess-3 code (i) 23, (ii) 15.46, and (iii) 643

Solution:

(i) 23

According to excess-3 code we need to add 3 to both digits in the decimal number then convert into 4-bit binary number for result of each digit. Therefore, 23+33=56=0101 0110 which is required Excess-3 code for given decimal number 23.

(ii) 15.46

According to Excess-3 code we need to add 3 to both digits in the decimal number then convert into 4-bit binary number for result of each digit. Therefore,15.46+33.33 = 48.79 = 0100 1000.0111 1001 which is required excess-3 code for given decimal number 15.46.

(iii) 643

Decimal number	:	6	4	3
Add 3 to each bit	:	+3	+3	+3
Sum	:	9	7	6

By converting the above sum into its equivalent binary code, we have

Sum ⟶	9	7	6
	↓	↓	↓
	1001	0111	0110

Hence, the excess-3 code of $(643)_{10}$ is 1001 0111 0110

Excess-3 CODE:

Solved Problem-5: Convert $(11110)_2$ to Excess-3 using binary

Solution:

Binary Number: $(11110)_2$

(i) Finding decimal equivalent of the number

$$(11110)_2 = ((1 \times 2^4) + (1 \times 2^3) + (1 \times 2^2) + (1 \times 2^1) + (0 \times 2^0))_{10}$$

$$= (16 + 8 + 4 + 2 + 0)_{10} = (30)_{10}$$

Decimal number of the Binary number $(11110)_2$ is $(30)_{10}$

(ii) Now, we add 3 in each digit of the decimal number

The decimal number is 30. Now, we will add 3 into the decimal number 30.

$= 30 + 33 = 63$

(iii) Now, we find the binary code of each digit of the decimal number 63

$$(63)_{10} = (0110)_2\ (0011)_2 = (01100011)_{\text{Excess-3}}$$

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Excess-3 CODE:

Advantages :

- These are unweighted binary decimal codes
- These are self complementary codes
- These codes use biased representation

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Gray code or cyclic code

The code which exhibits only a single bit change from one number to the next number is known as gray code or cyclic code. In this code, between any two successive code words, there will be change in only one position. This code is also called as unit distance code. It is also called as reflected code, because n-bit gray code can be generated by reflecting the $(n-1)^{th}$ bit code.

Table (8.5): *Grey code representation*

Decimal Code	BCD Code	Gray Code
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101

Gray code or cyclic code

Procedure to generate Gray code: The prefix and reflect method are recursively used to generate the gray code of a number. For generating gray code:

(i) We find the number of bits required to represent a number.

(ii) Next, we find the code for 0, i.e., 0000, which is the same as binary.

(iii) Now, we take the previous code, i.e., 0000, and change the most significant bit of it.

(iv) We perform this process reclusively until all the codes are not uniquely identified.

(v) If by changing the most significant bit, we find the same code obtained previously, then the second most significant bit will be changed, and so on.

Advantages:

- Better for reducing errors while converting analog to digital signals.
- Reduces the occurrence of "hamming walls" (an undesirable state) when used in genetic algorithms

Gray code or cyclic code

- It can be used to reduce the size of a logic circuit.
- Useful in clock domain crossing

Disadvantages:

- Not suitable for arithmetic operations
- Outside of a few select situations, it has limited practical use.

Applications:

Gray codes are widely used to prevent spurious output from electromechanical switches and to facilitate error correction in digital communications such as digital terrestrial television and some cable TV systems.

Gray to Binary code conversion

Conversion from gray to binary code can be achieved using the following seps:

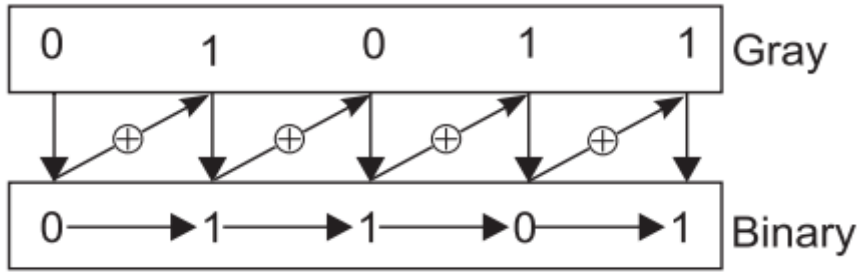
(i) The most significant bit(MSB) of the binary code is always equal to the MSB of the given gray code .

(ii) The 2nd bit of the binary number is the same as the 1st bit of the binary number when the 2nd bit of the gray code is 0;otherwise,the 2nd bit is altered bit of the 1st bit of binary number.it means if the 1st bit of the binary is 1,then the 2nd bit is 0, and if it is 0, then the 2nd bit be 1

(iii) Step 2 is repeated for each successive bit .

Gray to Binary code conversion

- Example: the binary number of the gray code 01011 is 01101
- The 1st bit of the binary number is the same as the MSB of the gray code.The MSB of the gray code is0,so the MSB of the binary number is 0.
- Now,for the 2nd bit ,we check the 2nd bit of the gray code. the 2nd bit if the gray code is 1,so the 2nd bit of the binary number is one that is altered number of 1st .
- The next bit of the gray code is 0;the 3rd bit is the same as the 2nd bit of the gray code,i.e.,1.
- The 4th bit of the gray code is 1;the 4th bit of the binary number is 0 that is the altered number of the 3rd bit .
- The 5th bit of the gray code is 1; the 5th bit of the binary number is 1; that is the altered number of the 4th bit of the binary number.
- So, the binary number of the gray code 01011 is 01101

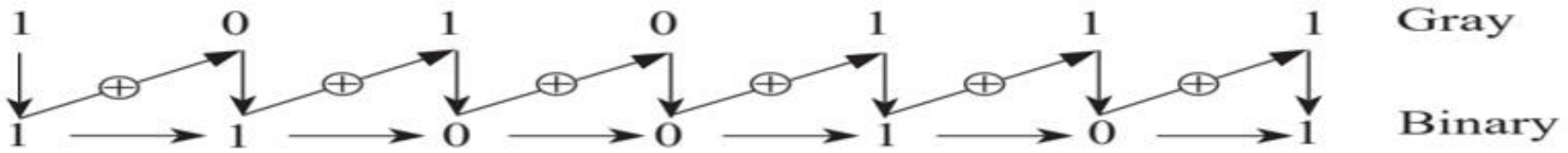


Gray to Binary code conversion

Solved Problem-6: Convert the following Gray codes into binary form.
(i) 1010111, (ii) 1011, and (iii) 101101

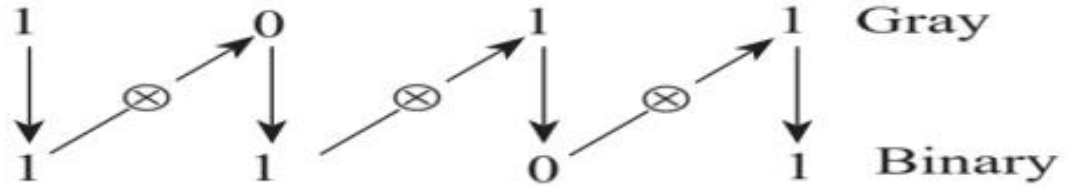
Solution:

(i) 1010111



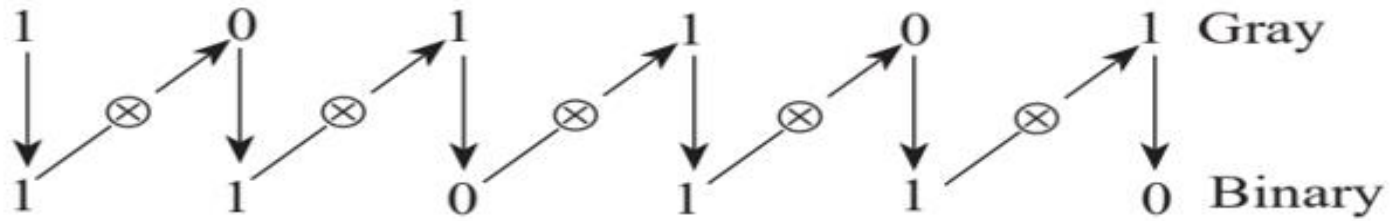
Therefore, $(1010111)_G = (1100101)_2$

(ii) 1011



Therefore, $(1011)_G = (1101)_2$

(iii) 101101



Therefore, $(101101)_G = (110110)_2$

Binary to Gray code conversion

Conversion from binary to gray code can be achieved using the following seps:

(i) The most significant bit(MSB) of the gray code is always equal to the MSB of the given binary number .

(ii) The second bit of the gray code equals to the exclusive-OR of the first and second bits of the binary number ,i.e., it will be 1 if these binary code bits are different and 0 if they are the same.

(iii) The third gray code bit is equals to the exclusive –OR of the second and three bits of the binary number, and so on……

Binary to Gray code conversion

Example: The gray code of the binary number 01101 is 01011

The 1st bit of the gray code is the same as the MSB of the binary number. The MSB is 0,so the MSB or 1st bit of the gray code is 0.

The XOR operation of the 1st and the second binary number.The 1st bit is 0,and the 2nd bit is 1.both the bits are different ,so the 2nd bit of the gray code is1 .

Now, perform the XOR of the 2nd bit and 3rd bit of the binary number.The 2nd bit is 1,and the 3rd bit is also 1.These bits are the same ,so the 3rd bit of the gray code is 0.

Again perform the XOR operation of the 3rd and 4th bit of the binary number.The 3rd bit is 1, and the 4th bit is 0.as these are different, the 4th bit of the gray code is 1.

Lastly ,perform the XOR of the 4th bit and 5th bit of the binary number.The 4th bit is 0, and the 5 th bit is 1.Both the bits are different ,so that the 5th bit of the gray code is 1.

The gray code of the binary number 01101 is 01011

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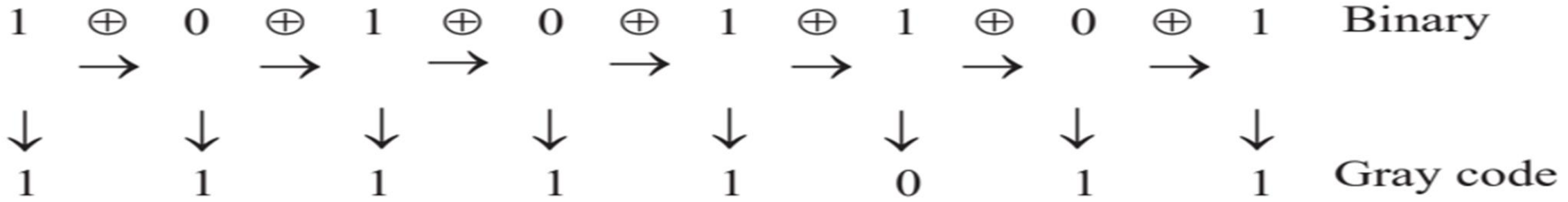
Binary to Gray code conversion

Solved Problem-7: Convert the following binary numbers into Gray codes.

(i) 10101101, and (ii) 110010

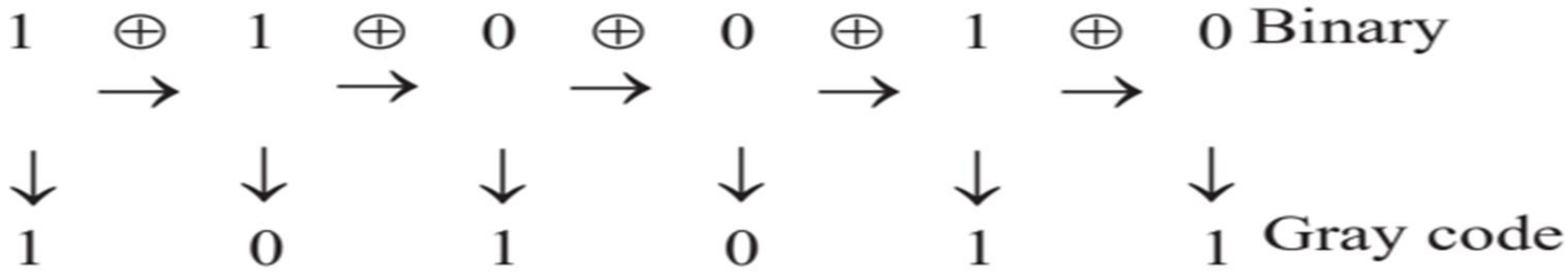
Solution:

(i) 10101101



Therefore, $(10101101)_2 = (11111011)_G$

(ii) 110010



Therefore, $(110010)_2 = (101011)_G$

Hamming Code

- Hamming code is a type of error –correcting code used in digital electronics and information theory to detect and correct errors that can occur during data transmission or storage. **Hamming code is a block code that is capable of detecting up to two simultaneous bit errors and correcting single-bit errors**. It was developed by R.W. Hamming for error correction.
- In this coding method, the source encodes the message by inserting redundant or parity bits within the message. These redundant bits are extra bits that are generated and inserted at specific positions in the message itself to enable error detection and correction. When the destination receives this message, it performs recalculations to detect errors and find the bit position that has error.

Hamming Code-Encoding

- The procedure used by the sender to encode the message encompasses the following steps –

✓ Step 1 – Calculation of the number of redundant or parity bits.

✓ Step 2 – Positioning the redundant or parity bits.

✓ Step 3 – Calculating the values of each redundant or parity bit.

Hamming Code-Encoding

- Once the redundant bits are embedded within the message, this is sent to the user.

Step 1 – Calculation of the number of redundant or parity bits.

Let m be the number of information or message bits, then the redundant bits is determined from the following formula, $2^p \geq m+p+1$.

for example ,if 4-bit information is to be transmitted ,then m=4.The number of redundant bits is determined by the trial and error method .

Let p=2,we get $2^2 \geq 4+2+1$, 4 not greater than or egqul to 7.so lets choose another value of p=3

Let p=3,we get $2^3 \geq 4+3+1$, the equation satisfies the condition. So number of redundant bits, p=3.

Hamming Code-Encoding

➢ **Step 2 – Positioning the redundant bits.**

The redundant bits placed at bit positions of powers of 2, i.e. 1, 2, 4, 8, 16 etc. Thus the locations of message bits and redundant bits are m4,m3,m2,p4,m1,p2,p1.

➢ **Step 3 – Calculating the values of each redundant bit.**

The assigned redundant bits are called parity bits. Each parity will check certain other bits in the total code group .It is one with bit location table ,as shown below.

Bit Location	**7**	**6**	**5**	**4**	**3**	**2**	**1**
Bit Designation	m4	m3	m2	p3	m1	p2	P1
Binary representation	111	110	101	100	011	010	001
Information/Message bits	m4	m3	m2		m1		
Parity bits				p4		p2	p1

Hamming Code-Encoding

Parity bit p1 covers all message bits in positions whose binary representation has 1 in the least significant position (001, 011, 101, 111, etc.). Thus pl checks the bit in locations 1, 3, 5, 7, 9, 11, etc.

Parity bit p2 covers all data bits in positions whose binary representation has 1 in the second least significant position (010, 011, 110, 111, etc.). Thus p2 checks the bit in locations 2, 3, 6, 7,10,11 etc.

Parity bit p4 covers all data bits in positions whose binary representation has 1 in the third least significant position (100, 101, 110, 111, etc.). Thus p4 checks the bit in locations 4 to 7, 12–15, 20–23 etc.

Each parity bit checks the corresponding bit locations and assign the bit value as 1 or 0, so as to make the number of 1s even for even parity and odd for odd parity.

Hamming Code-Encoding

Solved Problem-8: Encode a binary word 11000 into the even parity hamming code.

Solution: Given, the number of message bits, m = 5. Let us take ,p=4, then $2^4 \geq 5+4+1$

The equation is satisfied and redundant bits are 4.

So, the length of the hamming code = m+p=5+4=9.

The redundant bits are placed at bit positions 1, 2, 4 and 8.

Bit Location	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**
Bit Designation	m5	p8	m4	m3	m2	p4	m1	p2	P1
Binary representation	1001	1000	0111	0110	0101	0100	0011	0010	0001
Information/Message bits	1		1	0	0		0		
Parity bits		p8				p4		p2	p1

Hamming Code-Encoding

For parity bit pl: Bit locations 3, 5, 7 and 9 have two 1s. To have even parity, pl must be 0.

For parity bit p2: Bit locations 3, 6, 7 have one 1. To have even parity, p2 must be 1.

For p4: Bit locations 5, 6, 7 have one 1. To have even parity, p4 must be 1.

For p8: Bit locations 8, 9 have one I. To have even parity, p8 must be 1.

Thus the encoded 9-bit hamming code is 111001010.

Hamming Code-Encoding

Solved Problem-9: Encode a binary word 1010 into the even parity hamming code.

Solution: Given, the number of message bits, m 4.

Let us take, p=3 Then, $2^3 \geq 4+3+1$.

The equation is satisfied and redundant bits are 3.

So, the length of the hamming code = m+p=4+3=7.

The redundant bits are placed at bit positions 1, 2, and 4.

Bit Location	**7**	**6**	**5**	**4**	**3**	**2**	**1**
Bit Designation	m4	m3	m2	p4	m1	p2	P1
Binary representation	0111	0110	0101	0100	0011	0010	0001
Information/Message bits	1	0	1		0		
Parity bits				p4		p2	p1

Hamming Code-Encoding

For parity bit pl: Bit locations 3, 5, and 7 have two 1s. To have even parity, pl must be 0.

For parity bit p2: Bit locations 3, 6, and 7 have one 1. To have even parity, p2 must be 1

For parity bit p4: Bit locations 5, 6, and 7 have two 1s. To have even parity, p4 must be 0.

Thus, the encoded 7-bit hamming code is 1010010.

Bit Location	**7**	**6**	**5**	**4**	**3**	**2**	**1**
Bit Designation	m4	m3	m2	p4	m1	p2	P1
Binary representation	0111	0110	0101	0100	0011	0010	0001
Information/Message bits	1	0	1		0		
Parity bits				p4		p2	p1

Hamming Code-Decoding

- The steps involved that detect and correct the error of the coded bits obtained are as follows:

(i) With the below expression check the number of parity bits present in the received coded bits. $2^p \geq m + p + 1$

(ii) Now, position the redundant bits accurately like the way in encoding that it must be placed at positions of powers of 2.

(iii) Lastly, go for a parity check according to an even or odd parity scheme used during transmission using data and redundant bits.

Hamming Code-Decoding

Solved Problem-10: Detect and correct the error in the received odd parity hamming code of (110001101).

Solution: Checking the parity bits

Bit Location	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**
Bit Designation	m5	p8	m4	m3	m2	p4	m1	p2	P1
Binary representation	1001	1000	0111	0110	0101	0100	0011	0010	0001
Received code	1	1	0	0	0	1	1	0	1

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Solution: Checking the parity bits

For p1: Check the locations 1, 3, 5, 7, 9. There is three 1s in this group, which is correct for odd parity. Hence the bit value for pl is 1.

For p2: Check the locations 2, 3, 6, 7. There is one 1 in this group, which is correct for odd parity. Hence the bit value for p2 is 0.

For p4: Check the locations 4, 5, 6, 7. There is one 1 in this group, which is correct for odd parity. Hence the bit value for p4 is 1.

For p8: Check the locations 8, 9. There are two 1s in this group, which is wrong for odd parity. Hence the bit value for p8 is 0.

The resultant binary word is (p8, p4, p2, p1) = 0101. It corresponds to the bit location 5 in the above table .the error is detected in the data bit m2. The error is 0 and it should be changed to 1. Thus the corrected code is 100011101.

Bit Location	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**
Bit Designation	m5	p8	m4	m3	m2	p4	m1	p2	P1
Binary representation	1001	1000	0111	0110	0101	0100	0011	0010	0001
Received code	1	**1**	0	0	**0**	1	1	0	1
Correct code	1	0	0	0	1	1	1	0	1

Hamming Code-Decoding

Solved Problem-11: Is errors are present in the received even parity hamming code of (1010000). If errors are present then correct it.

Bit Location	**7**	**6**	**5**	**4**	**3**	**2**	**1**
Bit Designation	m4	m3	m2	p4	m1	p2	P1
Binary representation	0111	0110	0101	0100	0011	0010	0001
Received Code	1	0	1	0	0	0	0

Checking the parity bits

For p1: Check the locations 1, 3, 5, 7. There are two Is in this group, which is correct for even parity. Hence the bit value for pl is 0.

For p2: Check the locations 2, 3, 6, 7. There is one 1 in this group, which is wrong even parity. Hence the bit value for p2 is 1.

Hamming Code-Decoding

For p4: Check the locations 4, 5, 6, 7. There are two 1s in this group, which is correct for even parity. Hence the bit value for p3 is 0.

The resultant binary word is (p3, p2, pl) = (010). It corresponds to the bit location 2 in the above table. The error is detected in the parity bit p2. The error is 0 and it should be changed to 1. Thus the corrected code is 1010010

Bit Location	**7**	**6**	**5**	**4**	**3**	**2**	**1**
Bit Designation	m4	m3	m2	p4	m1	p2	P1
Binary representation	0111	0110	0101	0100	0011	0010	0001
Received Code	1	0	1	0	0	**0**	0
Correct code	1	0	1	0	0	1	0

TRUTH TABLES AND FUNCTIONALITY OF LOGIC GATES

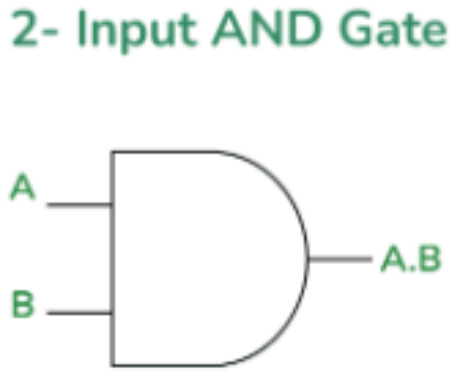
- Logic gates are the basic building blocks of any digital system.
- Logic gates have one or more than one input and only one output.
- Logic gates produces one output level based on different combinations of inputs.
- There are three basic logic gates such as 'AND', 'OR' and "NOT".
- Any complex logic operations can be performed with the help of these basic logic gates.
- These basic logic gates can also be interconnected to get a variety of logical operations, which is called **logic design**.
- Logic gates are nothing but electronic circuits because logic gates are made-up of electronic devices (Metal oxide semiconductor devices) and their related components.

TRUTH TABLES AND FUNCTIONALITY OF LOGIC GATES

- These logic gates are available in different forms such as **small-scale Integrated circuits (SSI), Large Scale Integrated circuits (LSI) and Very Large Integrated circuits (VLSI).**
- The inputs and outputs of logic gates have two levels such as 'HIGH' level and 'LOW' level or simply '1' and '0'.
- A table which represents an output for every possible combination of input variable is called **a truth table.**
- From the truth table, it can also say that how the output of a logic circuit is changing with respect to the various combinations of logic levels at the inputs.

AND Gate

- Figure 8.2(a) shows the logic symbol of the 2-input AND gate.
- Figure 8.2(b) shows that truth table of 2-input AND gate.
- An AND gate may have two or more inputs but only one output.
- For 2-input AND gate, there will be four possible combinations ($2^2 = 4$) of inputs such as '00', '01', '10', and '11'.



Truth Table

A (Input 1)	B (Input 2)	X = (A.B)
0	0	0
0	1	0
1	0	0
1	1	1

AND Gate With Truth Table

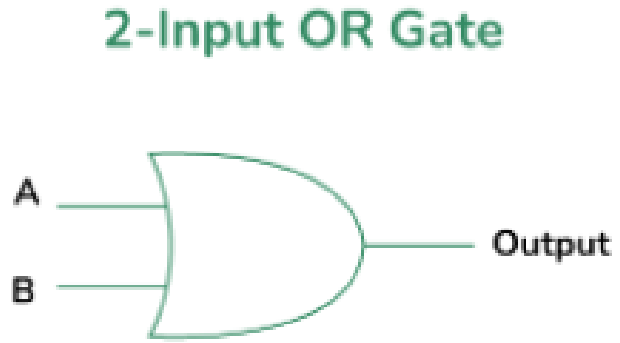
AND Gate

- For each combination of input variables, the output can be calculated with the help of output expression Y = AB.
- If the input variables are 'A' and 'B' then the output expression can be represented by Y = A.B, which reads as Y is equal to A and B or 'Y' is equal to AB or 'Y' is equal to A dot B.
- The symbol for the 'AND' operation is '.'.
- From the truth table, it can say that for the input combination '11' (A = 1 & B = 1), the output 'Y' is '1'.
- For the other input combinations '00', '01', and '10', the output 'Y' is '0'.
- Moreover, if one of the inputs is '0' then the output is '0'.
- If all the inputs are at 'HIGH' state or logic '1' state, then the output is '1'

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OR Gate

- Like an 'AND' gate, an 'OR' gate may have two or more inputs but only one output.
- Figure 8.3(a) shows that a logic symbol of two inputs 'OR' gate.
- Figure 8.3(b) shows that the truth table of two input 'OR' gate.
- For two inputs 'OR' gate, there will be four possible combinations ($2^2 = 4$) of inputs such as '00', '01', '10', and '11'.
- For each combination of input variables, the output can be calculated with the help of the output expression Y = A+B.



Truth Table

Input A	Input B	Output
0	0	0
0	1	1
1	0	1
1	1	1

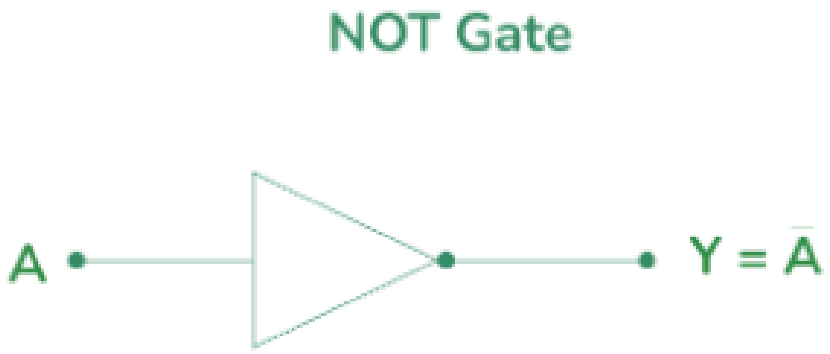
2 Input OR Gate

OR Gate

- The symbol for the 'OR' operation is '+'.
- If the input variables are 'A' and 'B' then the output expression can be represented by Y = A + B.
- This is read as Y is equal to A plus B.
- From the truth table, it may be noted that, for the input combination '00' (A = 0 & B = 0), the output is '0'.
- For the other combinations of inputs '01', '10', and '11', the output Y is '1'.
- Moreover, if one of the inputs is at a HIGH level, then the output is '1'.
- If all the inputs are at LOW state (0), then the output is '0'.

NOT Gate

- 'NOT' gate is also called an inverter.
- It has only one input and one output.
- The output of an inverter is always the complement of an inverter input.
- The symbol for 'NOT' operation is '-' (bar).



Truth Table

A (Input)	Y = Ā (Output)
0	1
1	0

NOT Gate With Truth Table

NOT Gate

- For a 'NOT' gate or inverter, if the input is 'A', then the output can be represented by the expression as Y = $\bar{A}$.
- This is read as 'Y' is equal to 'A bar'.
- From the truth table, it is noted that, if the input A= '0' then the output Y = '1'.
- Similarly, if the input A = '1', then the output Y = '0'.

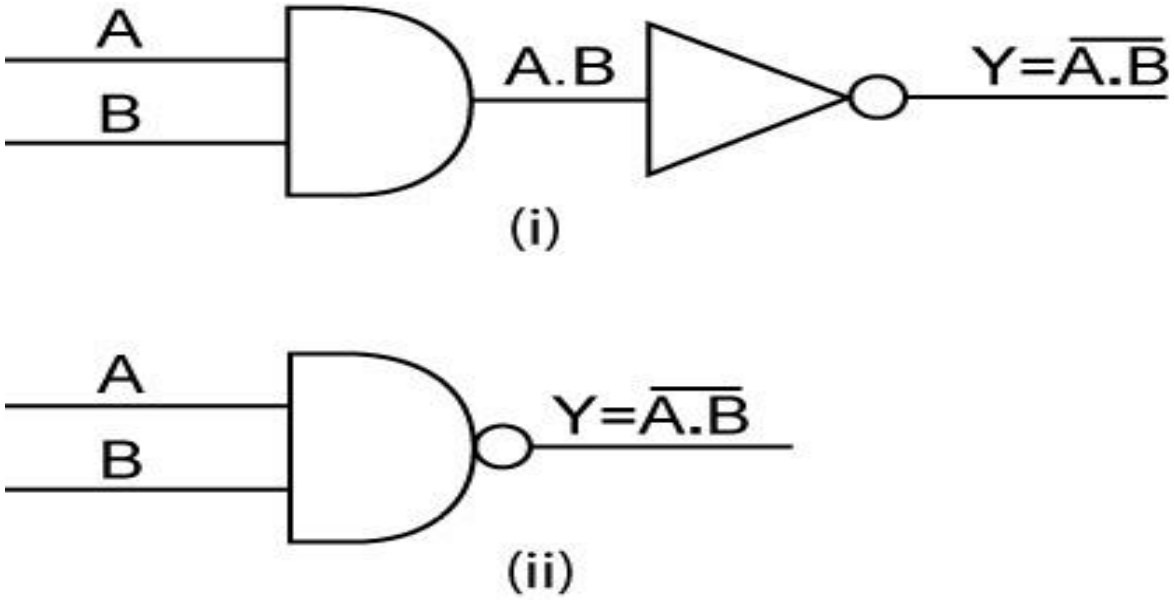
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The Universal Gates

- These basic gates are used to design any complexity of logic circuits.
- If the logic circuits use the three basic gates, then it is called as AND/OR/INVERT or AOI logic circuits.
- Even though we are using the three basic gates for any complexity of logic circuits, there are two universal gates such as 'NAND' and 'NOR'.
- The 'NAND' and 'NOR' gates are called universal gates because both 'NAND' and 'NOR' gates can perform all the three basic logic functions such as 'AND', 'OR' and 'NOT'.
- Therefore, AOI logic can be converted to 'NAND' logic or 'NOR' logic. In other words, we can say that any logic circuit can be designed by using either of 'NAND' gates or 'NOR' gates.
- In practice, 'NAND' and 'NOR' gates are economical and easier to fabricate than other logic gates.

NAND Gate

- The 'NAND' Gate is constructed by using an 'AND' gate and a 'NOT' gate.
- The 'NAND' gate is obtained by placing the 'NOT' gate at the output of the 'AND' gate.
- Figure 8.5(a) shows that an 'AND' gate is followed by a 'NOT' gate.
- Figure 8.5(b) shows that logic symbol of two-input NAND gate.



Input		output
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

NAND Gate

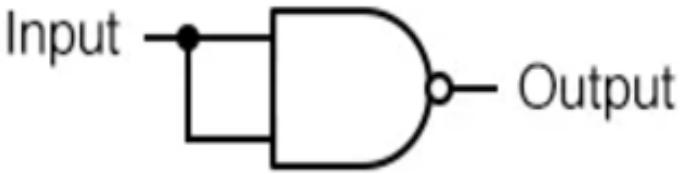
- The NAND gate may have two or more inputs but only one output.
- For two-input NAND gate, there will be four possible combinations of inputs ($2^2 = 4$) such as '00' (A = 0 & B = 0), '01', '10' and '11'.
- For each combination of input variables, the output Y can be calculated with the help of expression as $Y = \overline{A.B}$
- This output expression can be read as Y is equal to A.B whole bar or AB whole bar.
- From the truth table, it is noted that for the input combination of '11' (A = 1 & B = 1), the output Y is calculated as follows:
- A = 1, B = 1
- $Y = \overline{A.B}$
- We know that, $Y = \overline{1.1}$, $Y = \overline{1}$, Y=0 ,

NAND Gate

- Similarly, other combinations of input variables such as '00', '01', and '10', the output Y is '1'.
- From the truth table, it is also observed that if one of the input variable is '0' then the output Y is '1'.
- If all the input variables are at 'HIGH' state (logic 1 state), then the output Y is '0'.
- From the truth table of NAND gate, it can say that the output of NAND gate is equivalent to the complement of AND gate output.

NAND Gate as an Inverter

- To get NAND gate as an Inverter, tie up all the input terminals together as a single input or common terminal and apply it to a NAND gate to get a complementation of the common terminal.
- Figure (8.7) shows that two inputs NAND gate as an inverter.
- In figure (8.7), the two inputs are tied up as a single input and apply it to a 'NAND' gate to get a complementation of the input.

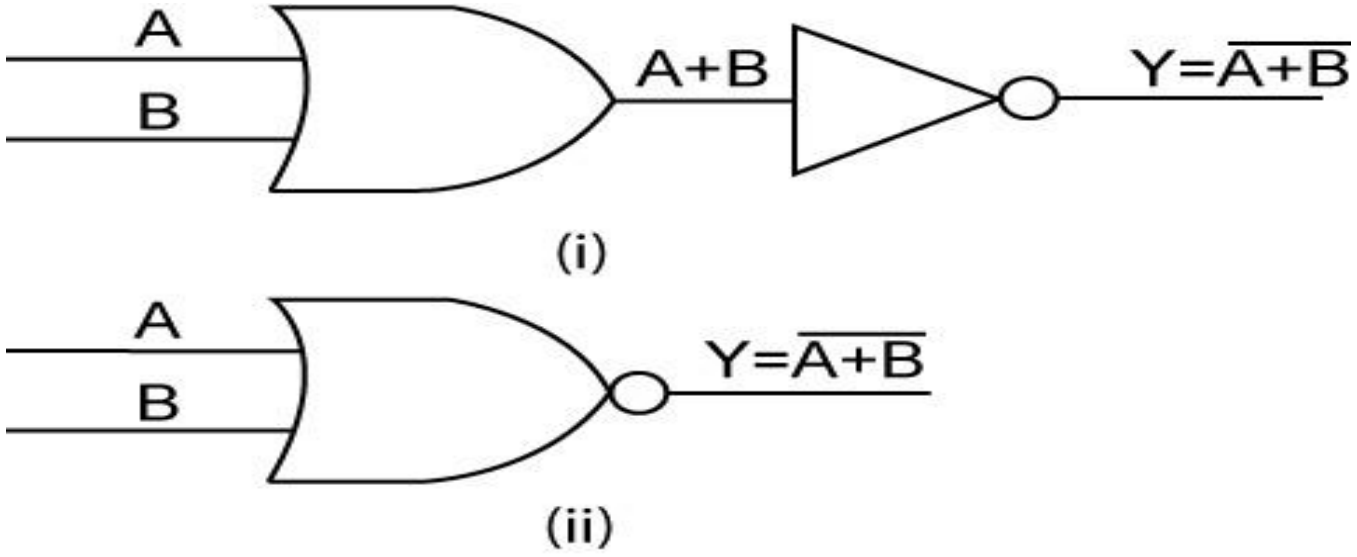




Input	Output
0	1
1	0

NOR Gate

- The NOR gate is constructed by using a 'OR' gate and a NOT gate.
- The NOR gate is obtained by placing a NOT gate at the output of a 'OR' gate.
- Figure 8.9(a) shows that a 'OR' gate is followed by a NOT gate.
- Figure 8.9(b) shows that logic symbol of two inputs NOR gate.



Input		output
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

NOR Gate

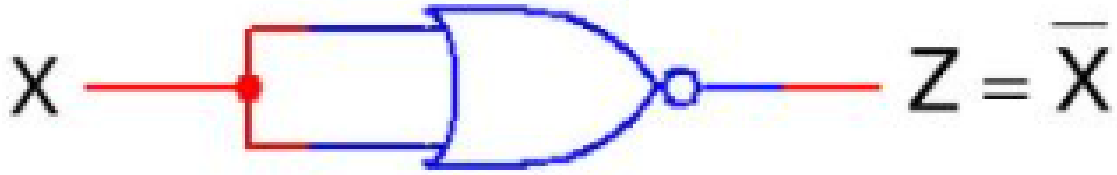
- The NOR gate may have two or more inputs but only one output. For two-input NOR gate, there will be four possible combinations of inputs such as '00' (A =0/ B = 0), '01', *10' and '11'.
- For each combination of input variables, the output 'Y' can be calculated with the help of an expression as Y = $\overline{A + B}$.
- This output expression can be read as 'Y' is equal to A + B whole bar.
- From the truth table, it is noted that, for the input combination of '00', the output Y is calculated as follows.
- Let the input combination = '00' A = 0, B = 0,Y = $\overline{A + B}$ = $\overline{0 + 0}$ = 1 [$\bar{0}$ = 1] ,Y = 1

NOR Gate

- Similarly, other combinations of input variables such as '01', '10 and '11', the output 'Y' is 0.
- From the truth table, it is observed that if one of the input variables is '1' then the output 'Y' is '0'.
- If all the input variables are at 'LOW' state (logic '0' state), then the output Y' is 1.
- From the truth table of the NOR gate, it can say that the output of the NOR gate is equivalent to the complement of the OR gate output.

NOR Gate as an Inverter

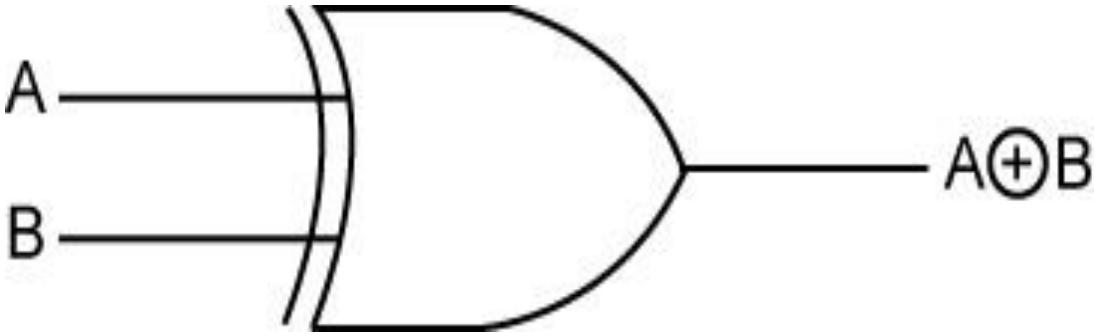
❖To get a NOR gate as an inverter, tie up all the input terminals together as a single input or common terminal and apply it to a NOR gate to get a complementation of the common terminal.



X	Z
0	1
1	0

Exclusive-OR(XOR) Gate

- The Exclusive-OR gate (or) XOR gate is mostly used in logic design.
- The XOR gate has two inputs and one output.
- **XOR gate operation for three or more than three variables does not exist**. If we want to get the XOR operation for three or more than three variables, a number of two input XOR gates will be used.



Input		output
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Exclusive-OR(XOR) Gate

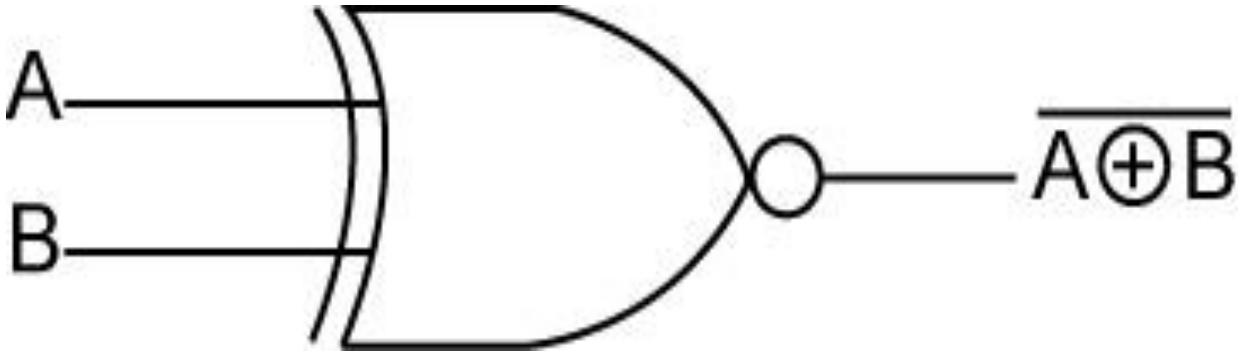
- If the input variables are 'A' and, 'B' then the output 'Y' can be expressed as, this is read as Y is ‘an exclusive OR B’.
- The symbol ‘⊕’ denotes 'Exclusive-OR' operation.
- The symbol ‘⊕’ is also called **modulo sum**.
- From the truth table of two inputs XOR gate, it can say that when the two input variables are different such as '01' and '10', then the output 'Y' will be '1’.
- The output ‘Y’ can be calculated for the above two input combinations as follows. Let the input combination=‘01’
- Now, A=0, B=1; $\mathbf{Y=A \oplus B}$; $\mathbf{Y=A.\bar{B} + \bar{A}.B}$ $=0.\bar{1} + \bar{0}.1 = 0.0 + 1.1 = 0+1=1$ $[\bar{0}=1, \bar{1}=0]$
- Similarly for input combination =‘10’
- Now, A=1, B=0; $\mathbf{Y=A \oplus B}$; $\mathbf{Y=A.\bar{B} + \bar{A}.B}$ $=1.\bar{0} + \bar{1}.0 = 1.1 + 0.0 = 1+0=1$; $[\bar{0}=1, \bar{1}=0]$

Exclusive-OR(XOR) Gate

- Moreover ,when the two input variables are equal such as '00' and '11', then the output 'Y' will be '0'.
- for the above two input combinations as follows. Let the input combination='00'
- Now, A=0, B=0; Y=A $\oplus$ B;Y=A. $\bar{B}$ + $\bar{A}$ .B =0. $\bar{0}$ + $\bar{0}$ .0 =0.1 +1.0=0+0=0 [$\bar{0}$ = 1, $\bar{1}$ =0]
- Similarly for input combination ='11'
- Now, A=1, B=1; Y=A $\oplus$ B;Y=A. $\bar{B}$ + $\bar{A}$ .B =1. $\bar{1}$ + $\bar{1}$ .1 =1.0 +0.1=0+0=0 ; [$\bar{0}$ = 1, $\bar{1}$ =0]

Exclusive-NOR(XNOR) Gate

- An exclusive-NOR gate or XNOR gate can be obtained by using XOR gate and a NOT gate.
- The XNOR gate is also mostly used in logic design.
- The XNOR gate has two inputs and one output. **The XNOR gate operation for three or more than three variables does not exist**.
- If we want to get the XNOR operation for three or more than three variables, a number of two inputs XNOR gates will be used.



Input		output
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Exclusive-NOR(XNOR) Gate

- If the input variables are 'A' and 'B' then the output 'Y' of the XNOR gate can be expressed as Y=AʘB.
- This is read as Y is "An exclusive-NOR 'B' .
- The output of 'XNOR' gate can also be written as Y= AB+ $\overline{AB}$.
- The symbol 'ʘ' denotes the "Exclusive-NOR' operation.
- From the truth table of two inputs XNOR gate, it may be noted that , when the two input variables are different such as '01' and '10', the output "Y" will be "0". The output "Y" can be calculated for the above two input combinations as follows: Let the input combination='01'
- Now, A=0, B=1; **Y=A ʘ B**;**Y=A. B+** $\overline{A.B}$ =0. 1+ $\bar{0}$. $\bar{1}$=0.1 +1.0=0+0=0 [$\bar{0}$ = 1, $\bar{1}$ =0]
- Similarly for input combination ='10'
- Now, A=1, B=0; **Y=A ʘ B**; **Y=A. B+** $\overline{A.B}$ =1. 0+ $\bar{0}$. $\bar{1}$ =1.0 +0.1=0+0=0 ; [$\bar{0}$ = 1, $\bar{1}$ =0]

Exclusive-NOR(XNOR) Gate

- Moreover, when the two input variables are equal such as '00' and '11', then the output 'Y' will be '1'.
- **From the truth table of two-input XNOR gate, it is noted that the output of XNOR gate is equivalent to the complement of the output of the XOR gate.**
- Hence, the output 'Y' of the XNOR gate can be written as follows.
- $\mathbf{Y = A \odot B = \overline{A \oplus B}}$

Basic Rules of Boolean Algebra

Basic Rules of Boolean Algebra

1. $A + 0 = A$	7. $A \cdot A = A$
2. $A + 1 = 1$	8. $A \cdot \overline{A} = 0$
3. $A \cdot 0 = 0$	9. $\overline{\overline{A}} = A$
4. $A \cdot 1 = A$	10. $A + AB = A$
5. $A + A = A$	11. $A + \overline{A}B = A + B$
6. $A + \overline{A} = 1$	12. $(A + B)(A + C) = A + BC$

DeMorgan's Theorem

$$\overline{(AB)} = (\overline{A} + \overline{B}) \qquad \overline{(A + B)} = (\overline{A}\ \overline{B})$$

Combinational Circuits

- Digital systems can be designed by using both combinational and sequential circuits .
- Combinational circuits can be designed by using logic gates which may be 'AND' 'OR' 'NOT' 'NAND' 'NOR' 'XOR' and 'XNOR'.
- Combinational circuits are used to perform a particular operation.
- The function of the combinational circuits is to take the input variables (1 or 0) and process them and give the output.
- In the combinational circuits, the output depends only on the present input variables.



Combinational Circuits

- As shown in the figure (8.14), there are 'n' input variables and 'm' output variables.
- For those 'n' input variables, there will be possible combinations of input variables.
- For each combination of input variables, there will be only one output combination.
- For example, let us consider two input variables that are present at the input side of the combinational circuit.
- For these two inputs, there will be 4 (2^2) possible combinations such as '00' '01' '10' and '11'. For each combination of input variables, there will be one output combination.



Design procedure for combinational circuit

To design a combinational circuit, the following procedure is used

(a) Obtain the problem definition.

(b) Determine the required input and output variables.

(c) Draw the truth table based on the relationship between input and output variables.

(d) Obtain the simplified Boolean function for each output.

(e) Finally, draw the logic diagram for the simplified Boolean function.

Adders

- As we know, Central Processing Unit (CPU) is the main building block of the digital computer. CPU performs a variety of operations. In the CPU, the Arithmetic and Logic Unit (ALU) is used to perform the arithmetic and logical operations.
- In arithmetic operations, addition is one operation which is used to add two binary digits. Addition of two binary digits can be done in four possible ways, which are given as follows.0+0=0,0+1=1,1+0=1,1+1=10.
- In the first three combinations, the sum output consists of only one digit, but in the fourth combination , the sum output consists of two digits.
- In these two digits (10) ,the highest significant bit (1), is called carry bit .
- Generally ,when we add the large number of bits, the carry will be generated and forwarded to the next higher significant bits.
- There are so many types of adders such as half-adder, full -adder, parallel binary adder,carry-look –ahead adder carry-skip adder and carry –save adder

The Half-Adder

A half-adder is a combinational digital circuit, which is used to add the two binary bits or digits. It also consists of two inputs and two outputs such as sum (S) and carry (C).



(a) Block diagram

Inputs		Outputs	
x	y	Sum(S)	Carry (C)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

(b) Truth table

Figure (8.15): *Half-adder*

Figures 8.15(a) and 8.15(b) shows the block diagram and a truth table of a half-adder. From the truth table, it is noted that the sum output is equivalent to the output of the two input 'XOR' gate. Hence, the sum output can be written as follows:

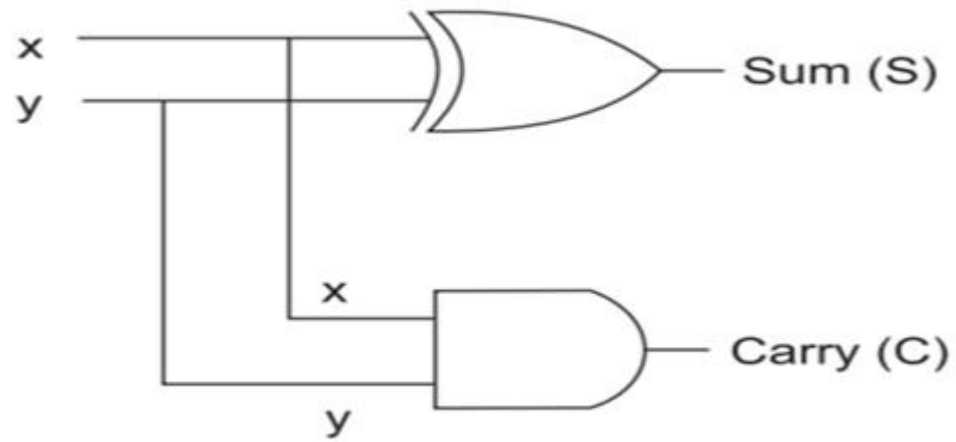
$$Sum(S) = x \oplus y$$

$$Sum(S) = x\,\bar{y} + y\,\bar{x} \qquad \ldots(8.5)$$

Similarly, the carry output is equivalent to the output of the two input 'AND' gate. Hence, the carry output can be written as

$$Carry\ (C) = xy \qquad \ldots(8.6)$$

A half-adder can be implemented in terms of logic gates by using the equation (8.5) and (8.6).



The Full-Adder

A full-adder is a combinational digital circuit, which is used to add the three binary bits or digits. A full-adder circuit consists of three inputs and two outputs such as sum(S) and carry (C). For the three inputs, there will be eight possible combinations and for each combination, there will be one output combination such as sum(S) and carry(C).

Inputs			Output	
x	*y*	*z*	Sum(S)	Carry(C)
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

(a) Truth table



(b) Block diagram

Figure (8.17): *Full-adder*

The Full-Adder

- Figures 8.17(a) and 8.17(b) shows the truth table and a block diagram of full adder.
- In the full adder, first we add the two bits, and then find the sum and carry.
- The carry bit is taken as carry output. Again the sum bit and the third bit will be added to get the final sum Output.
- For example, let us consider the input combination '110' (x = 1, y = 1& z = 0).
- For this combination, the first two bits are'11'.
- The sum and carry of these two bits are '0' and 1.
- Again, the sum bit (0) is added to the third bit '0'.
- After addition, the final sum output is '0'.

The Full-Adder

- Finally, the sum and carry bits for the input combination '110' are '0' and '1'.
- In this way, the outputs sum and carry will be obtained for all the possible combinations of three input variables.
- From the truth table of 8.17(a), the expressions for the sum and carry outputs can be written based on the input combinations for which the sum and carry outputs will be '1'.

The Full-Adder

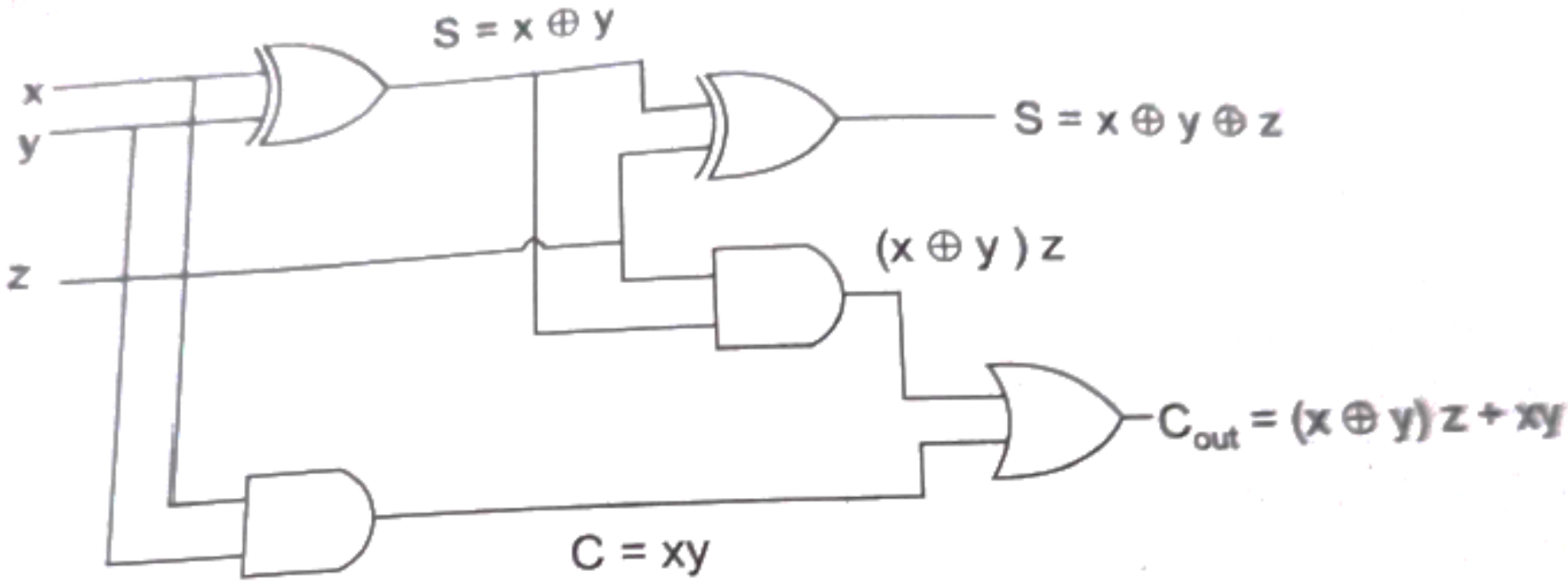


Figure (8.20): *Logic diagram of a full adder using two half-adder circuits*

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Thank You